

ECE/CS 757 Advanced Computer Architecture II

Spring 2025

Synchronization (Part 2)

Instructor: Prof. Joshua San Miguel

Some slides adopted from Greg Steffan and Natalie Enright Jerger

Last Lecture

- Mutual exclusion
- Test-and-set locks
- Ticket and queuing locks
- Deadlocks

This Lecture

- Centralized barriers
- Tree barriers
- Shared memory CMPs

Barriers

Necessary waiting in phase-based multithreaded applications.

Communication Overhead:

- Implementation cost of collective communication (e.g., multicast wakeup)
- Eliminate via caching and locality

Blocking Overhead:

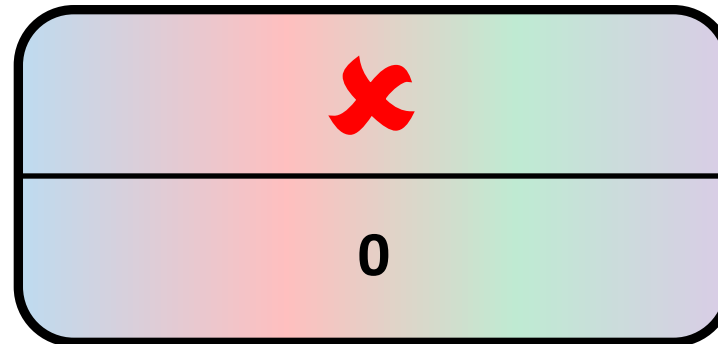
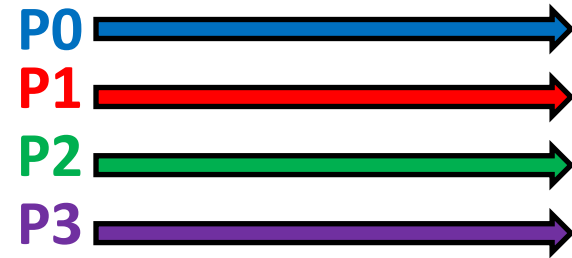
- Fundamental cost of waiting for slowest thread (i.e., load imbalance)
- Hide via speculative early wakeup

Barriers

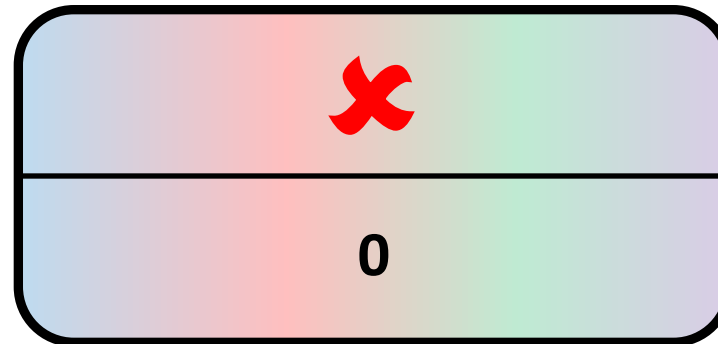
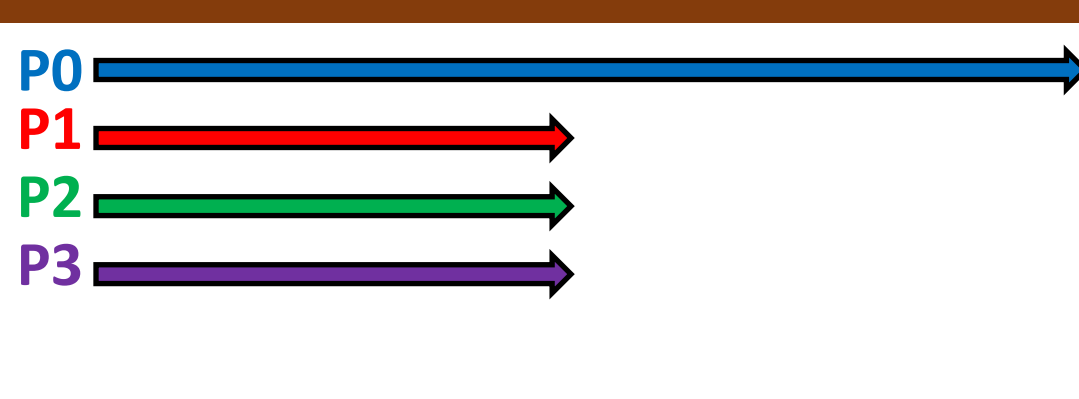
Design Objectives:

- Low critical path latency (i.e., from first thread arriving to last thread departing)
- Low serialization cost (i.e., fetch-and-add operations)
- Low interconnect congestion (i.e., minimize number of sharers per memory location)
- Low storage cost
- Scalability (all of the above in proportion to thread count)

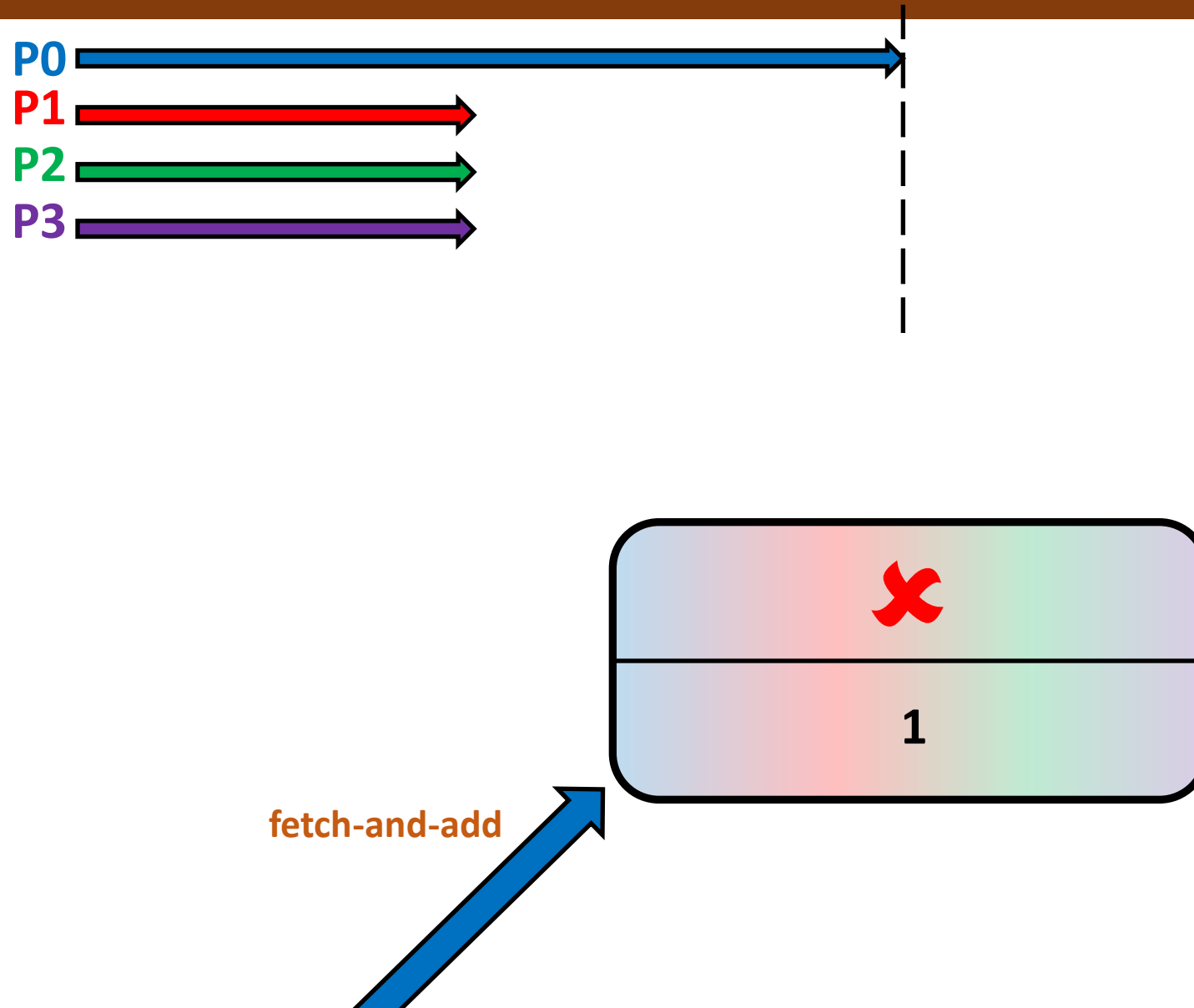
Centralized Barrier



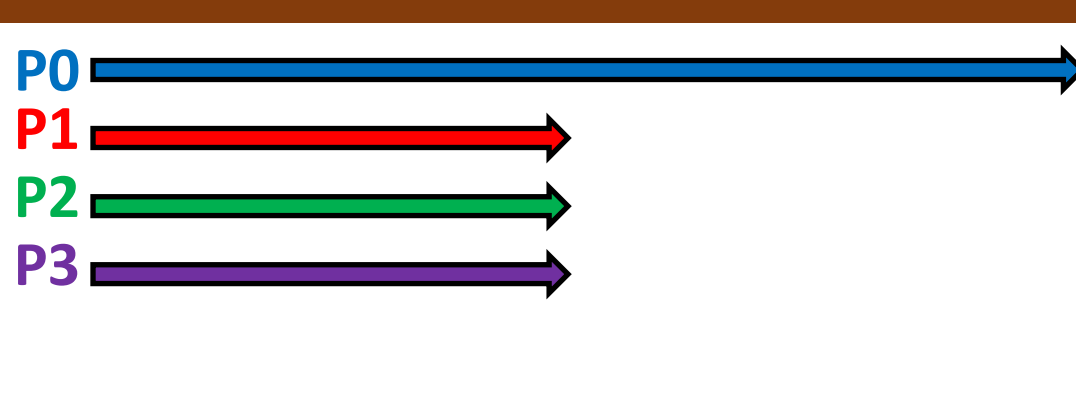
Centralized Barrier



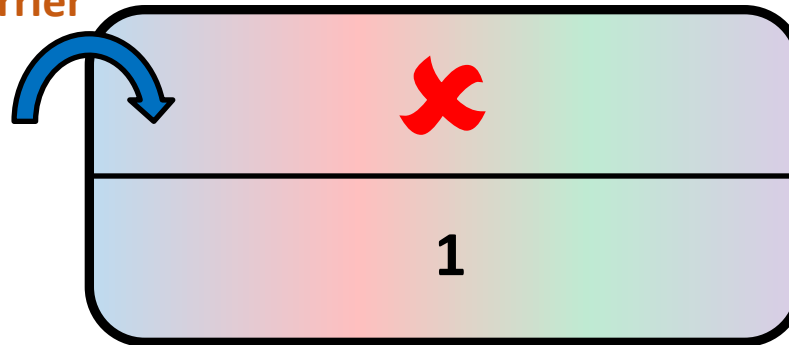
Centralized Barrier



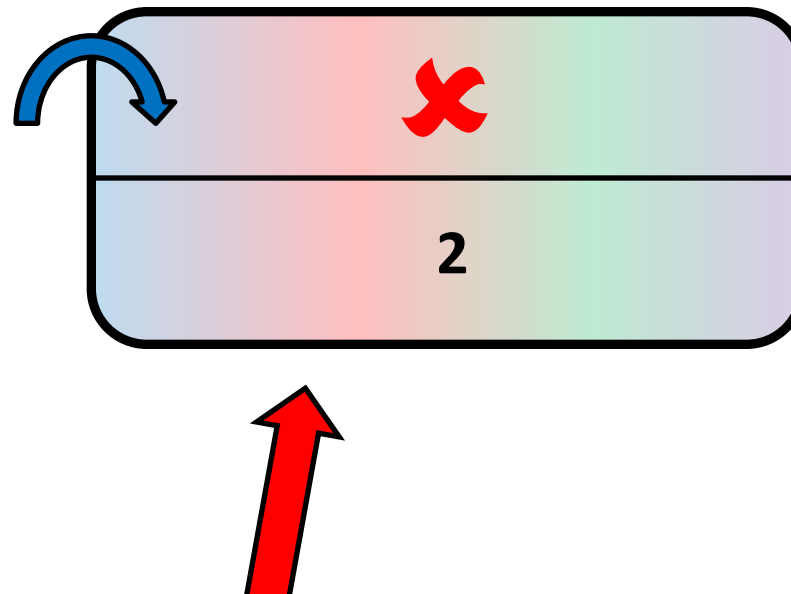
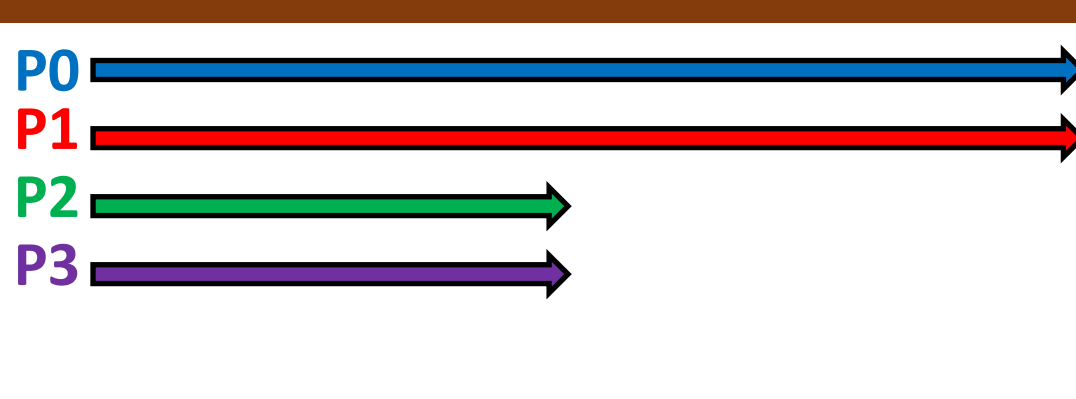
Centralized Barrier



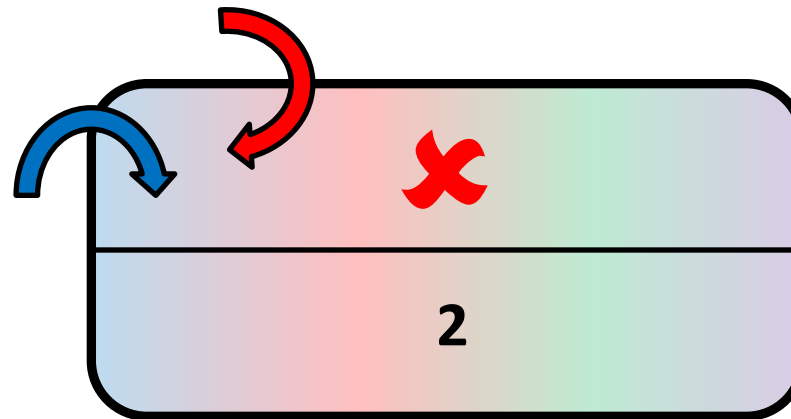
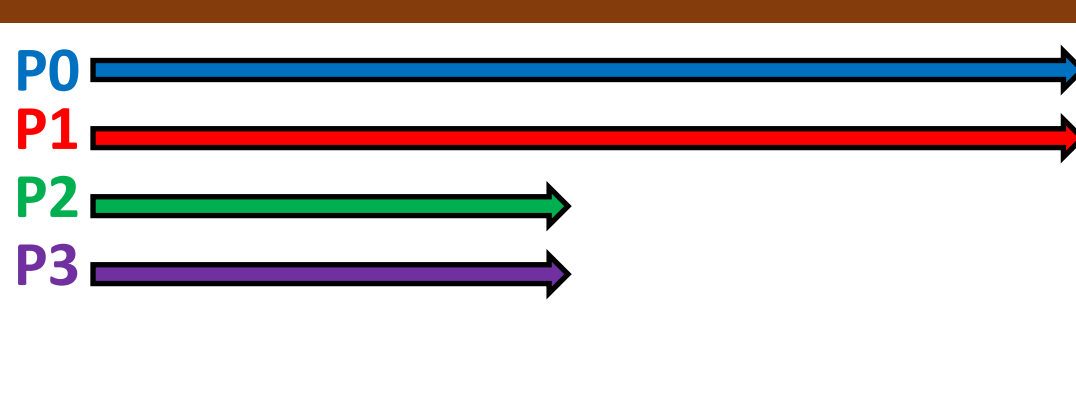
spin on barrier



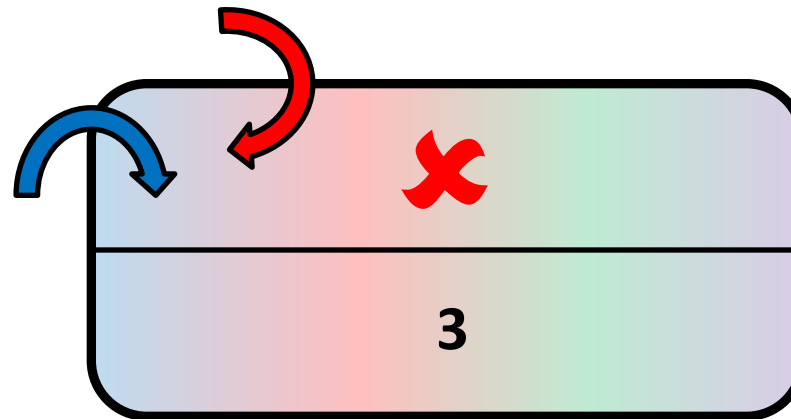
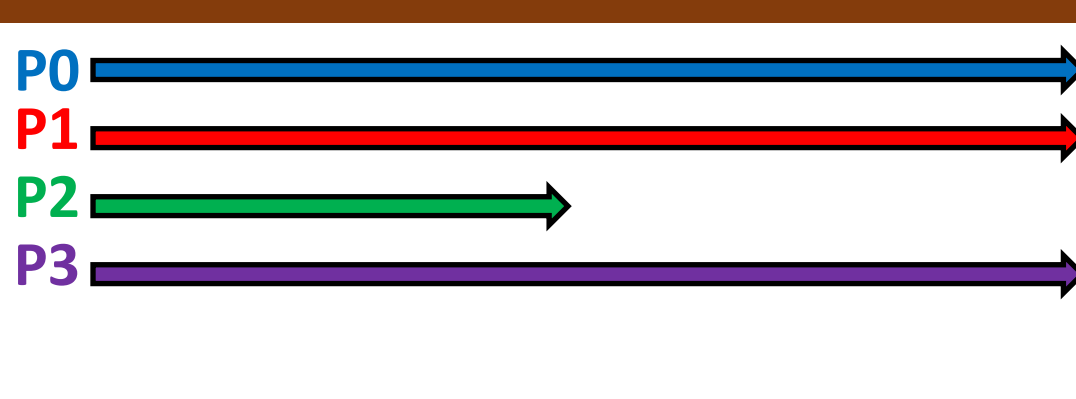
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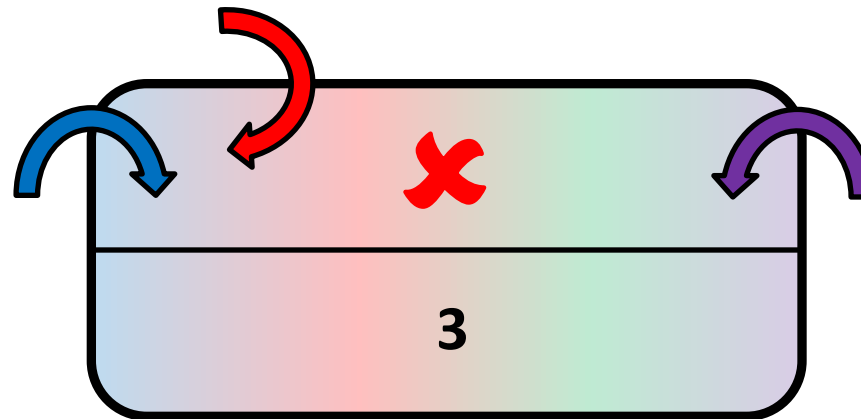
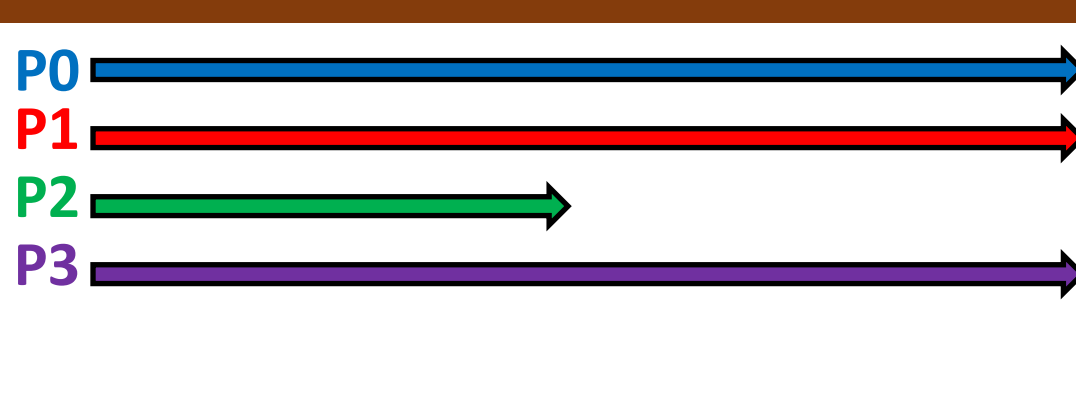
Centralized Barrier



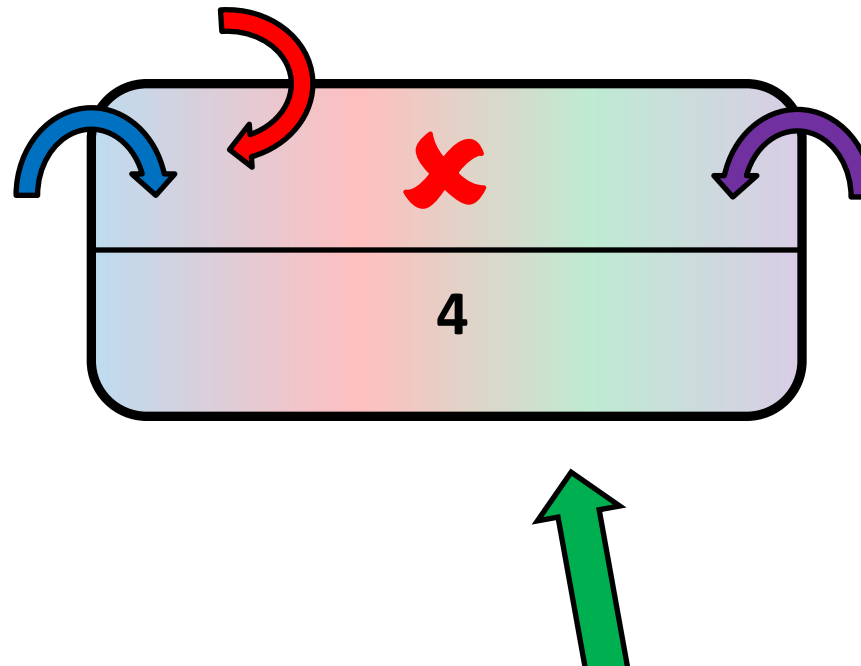
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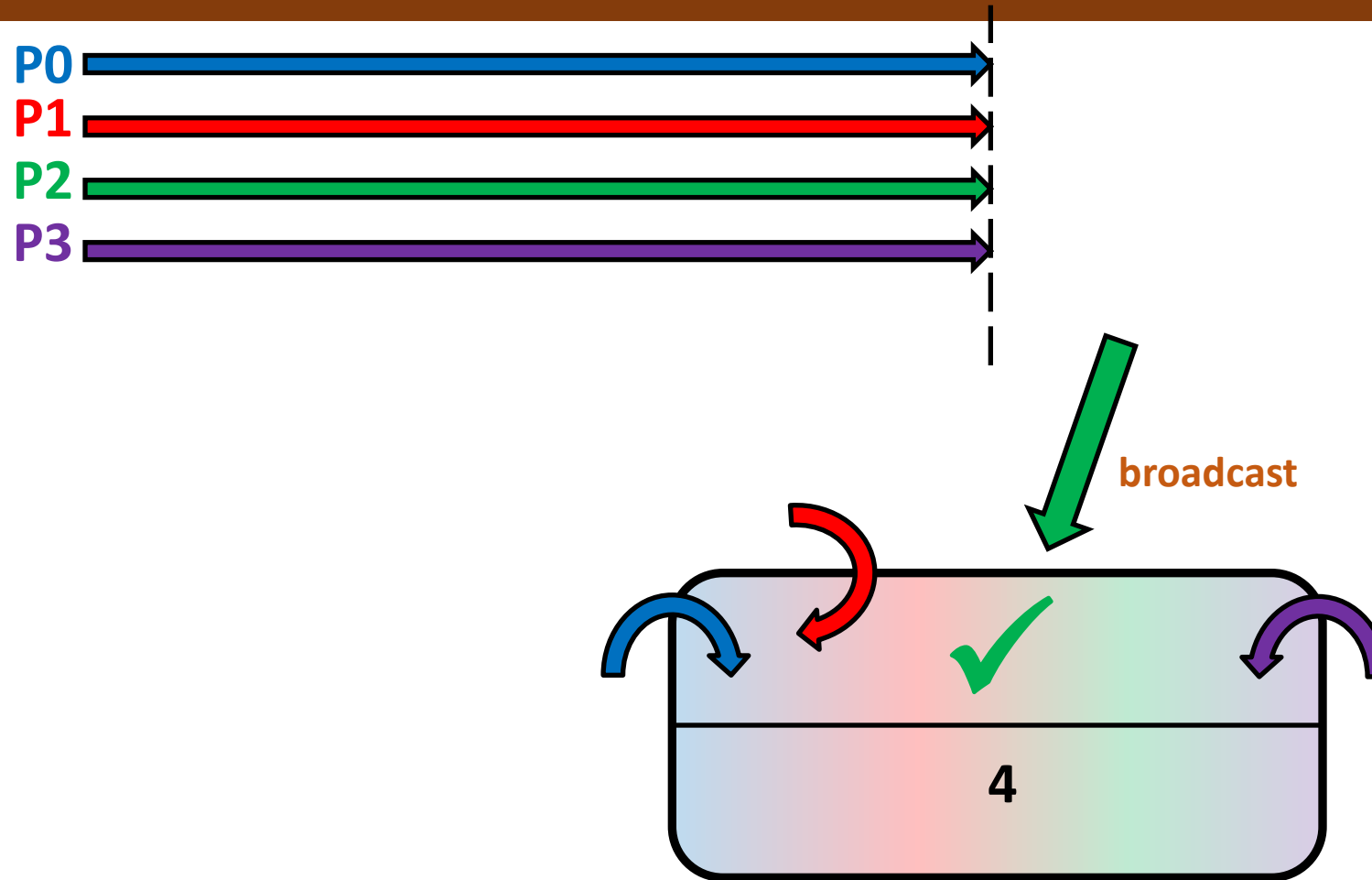
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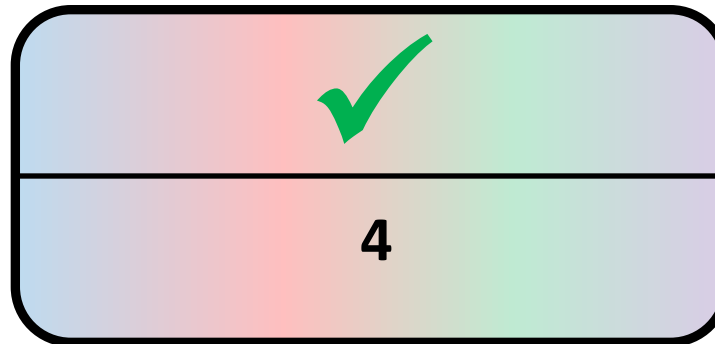
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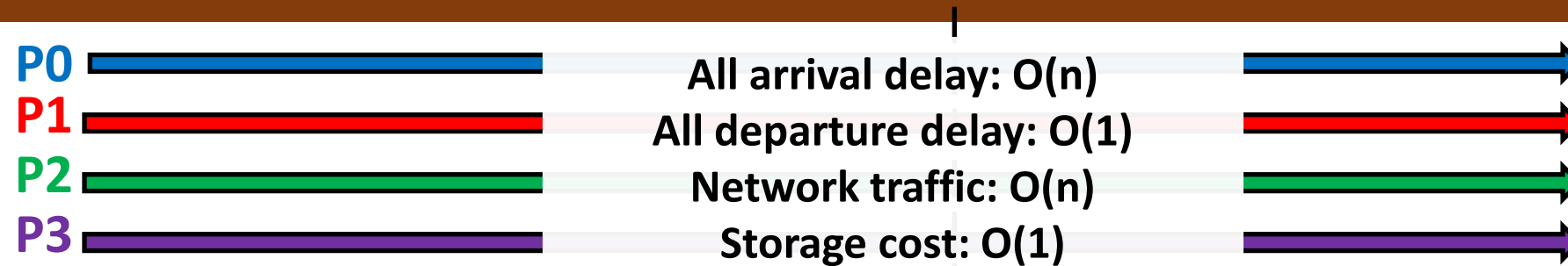
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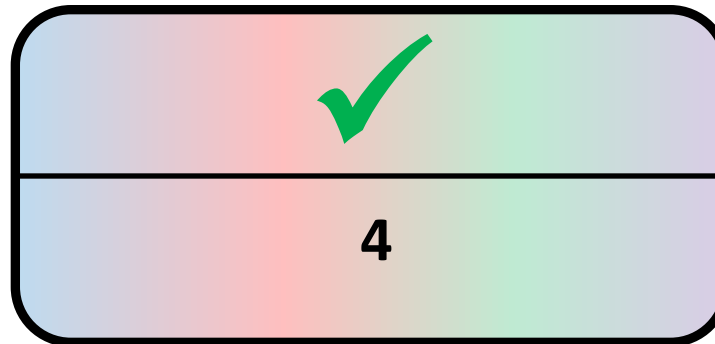
Centralized Barrier



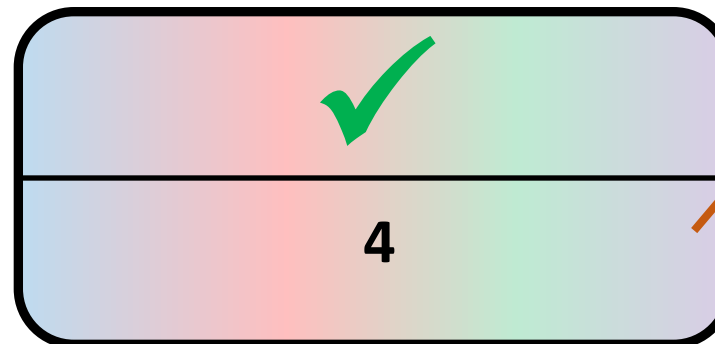
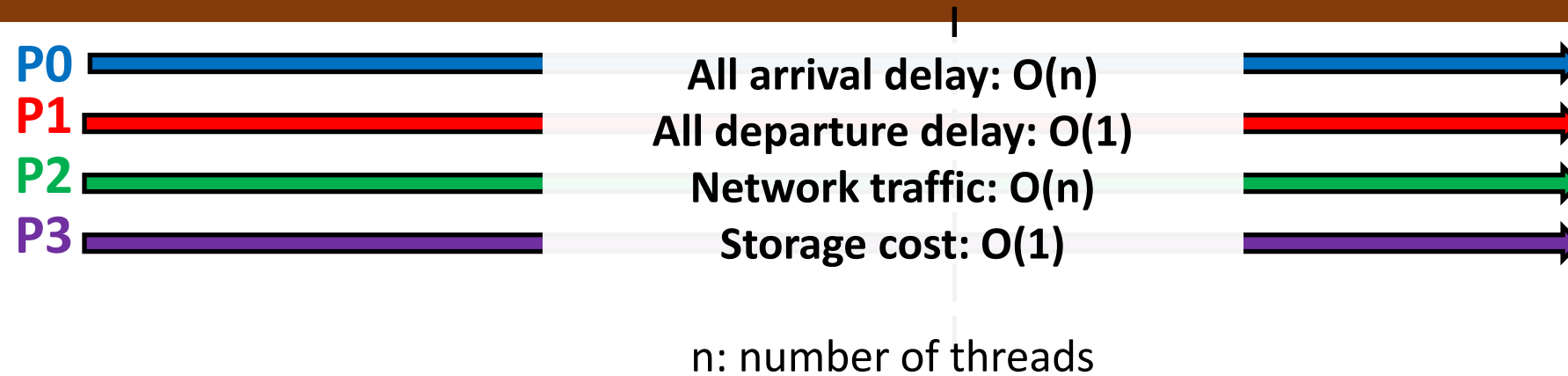
Centralized Barrier



n : number of threads



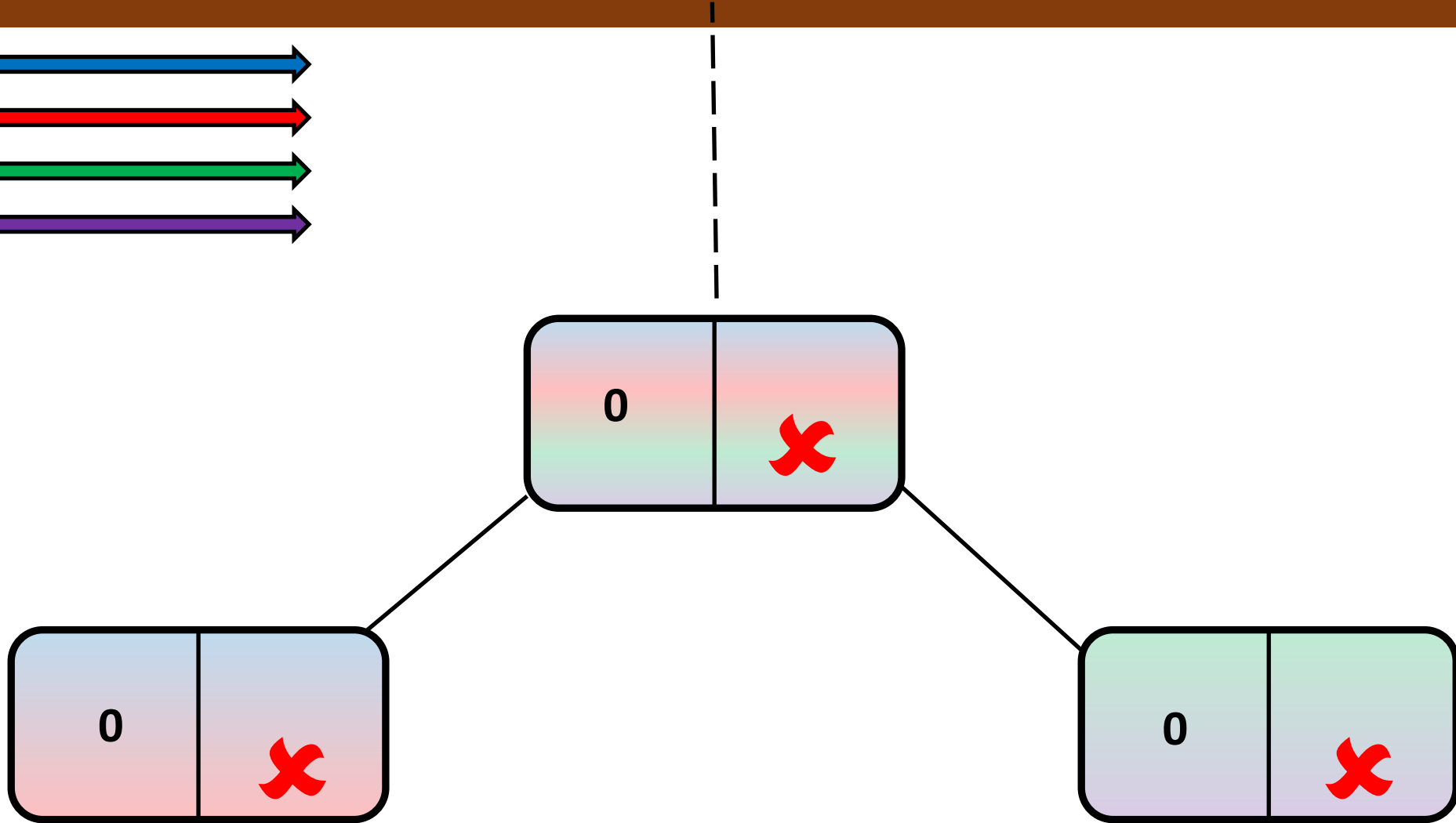
Centralized Barrier



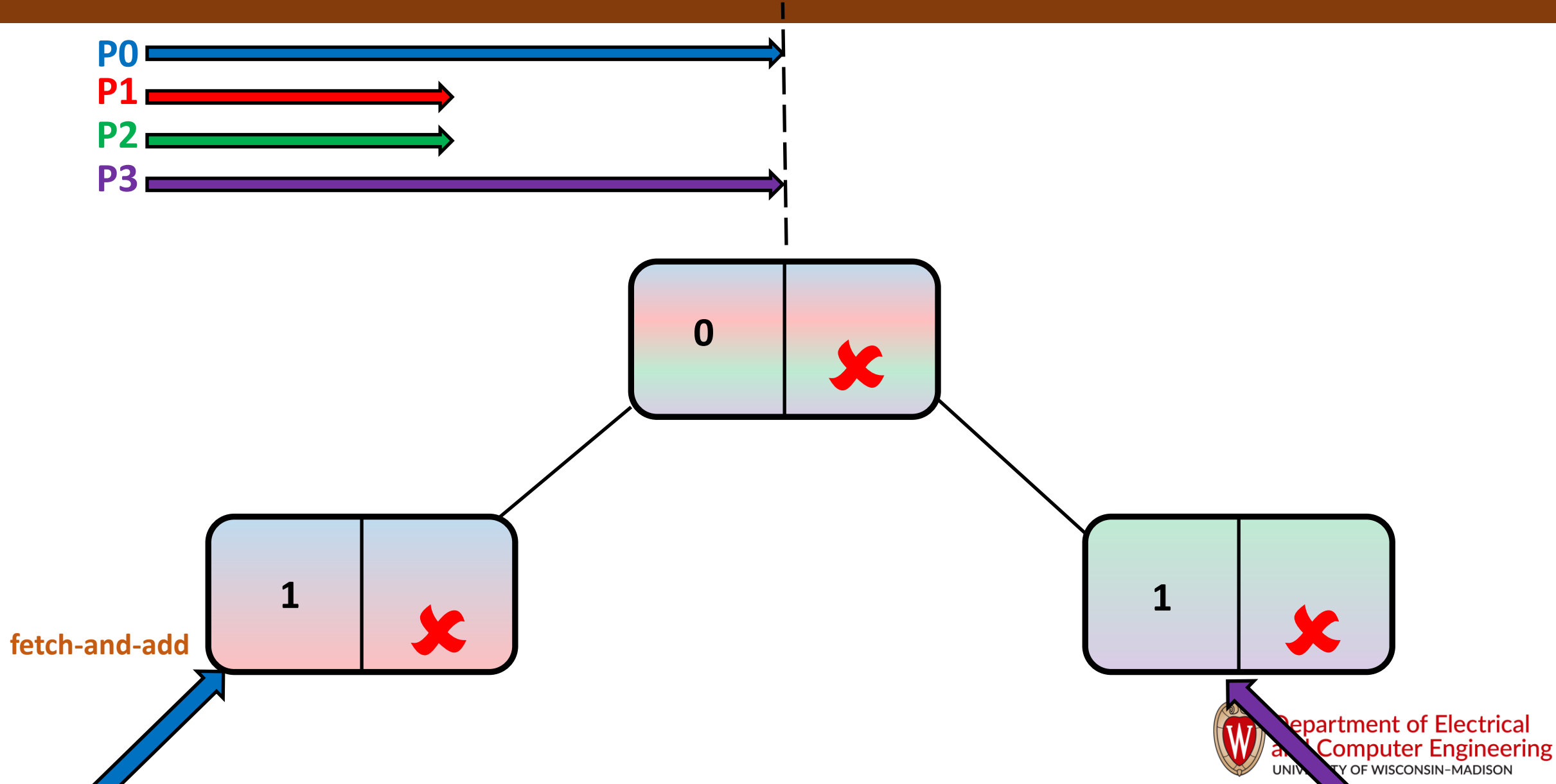
all threads fetch-and-add on
same memory location!

Software Combining Tree Barrier

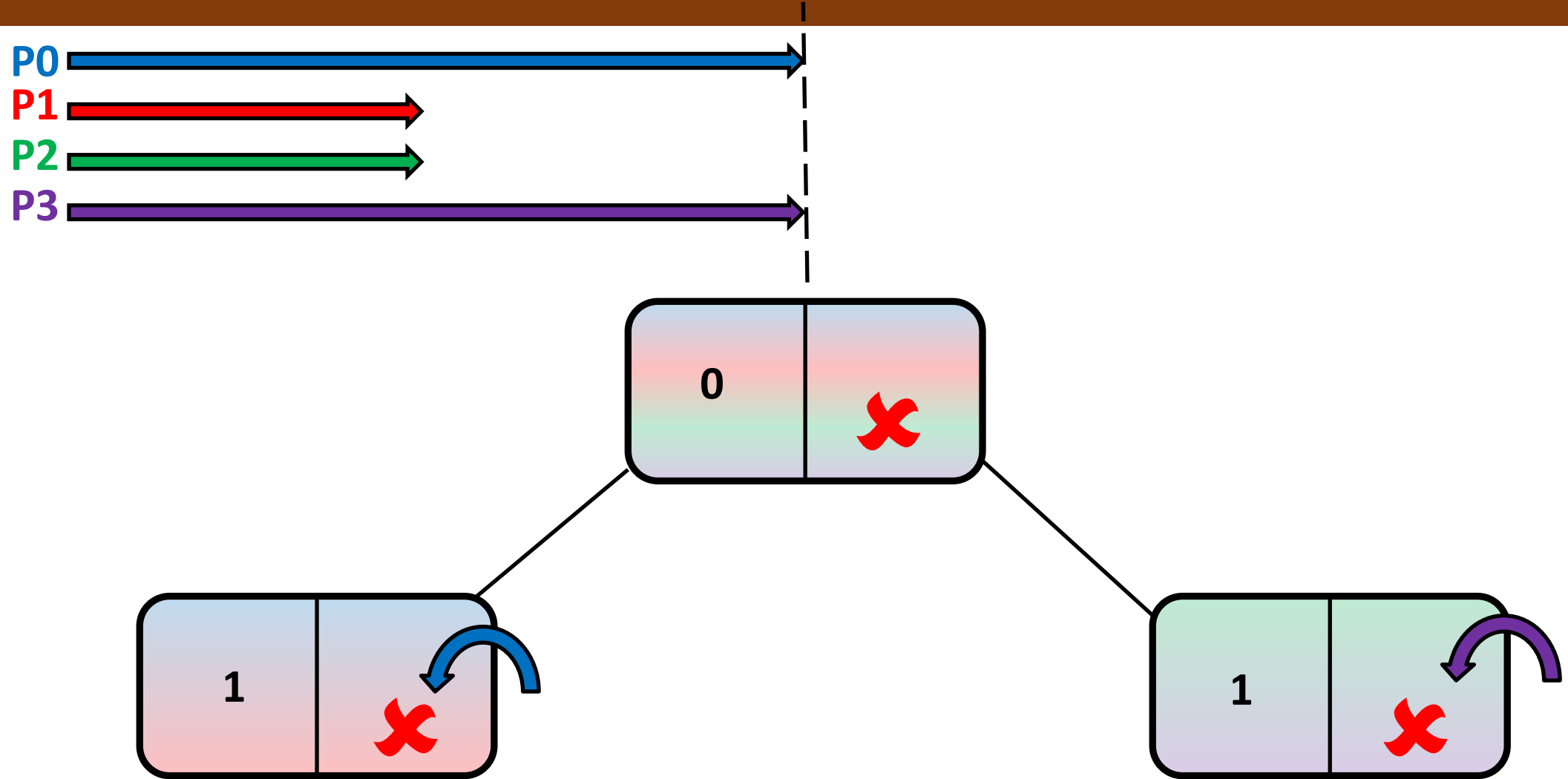
P0 →
P1 →
P2 →
P3 →



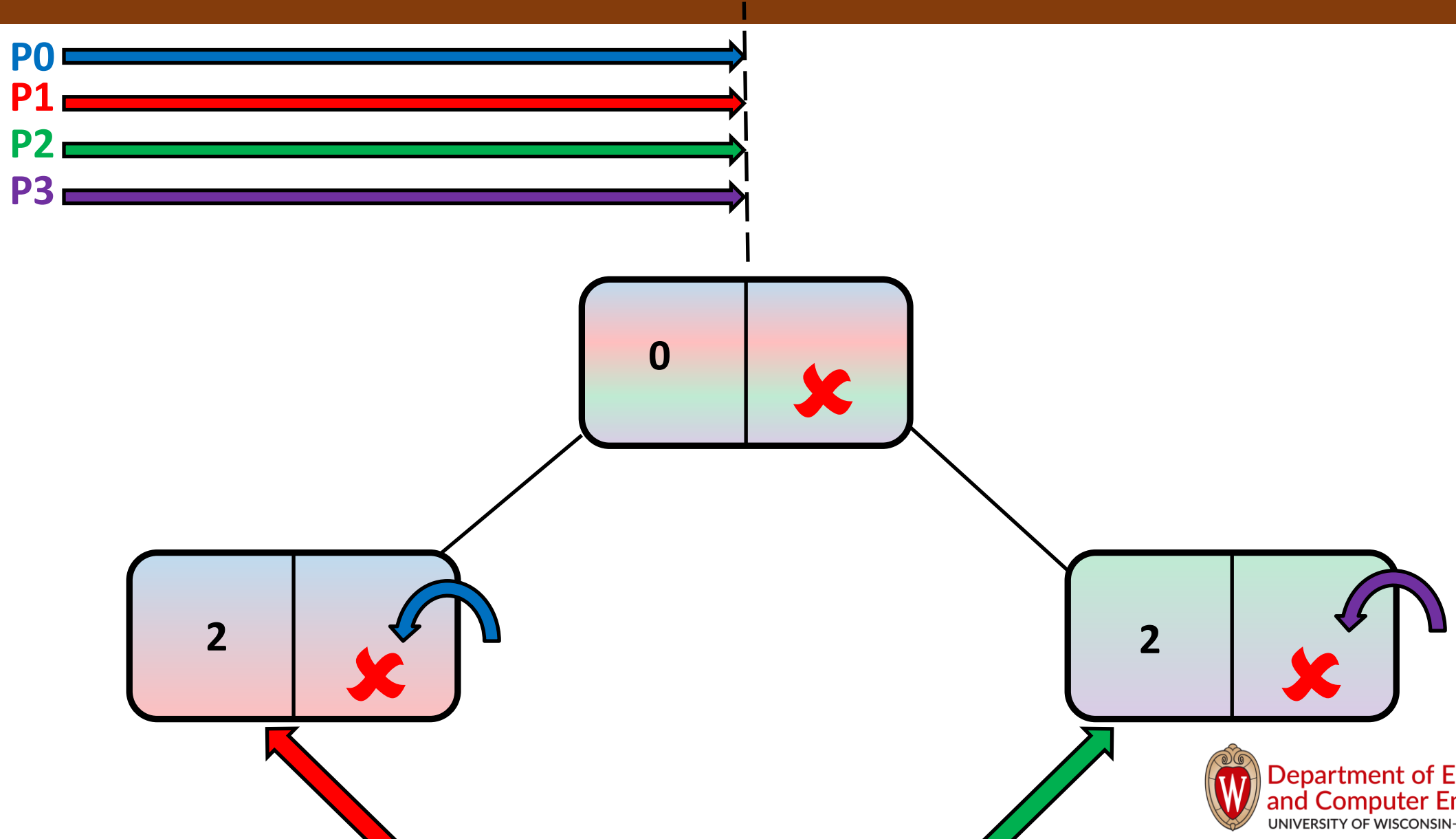
Software Combining Tree Barrier



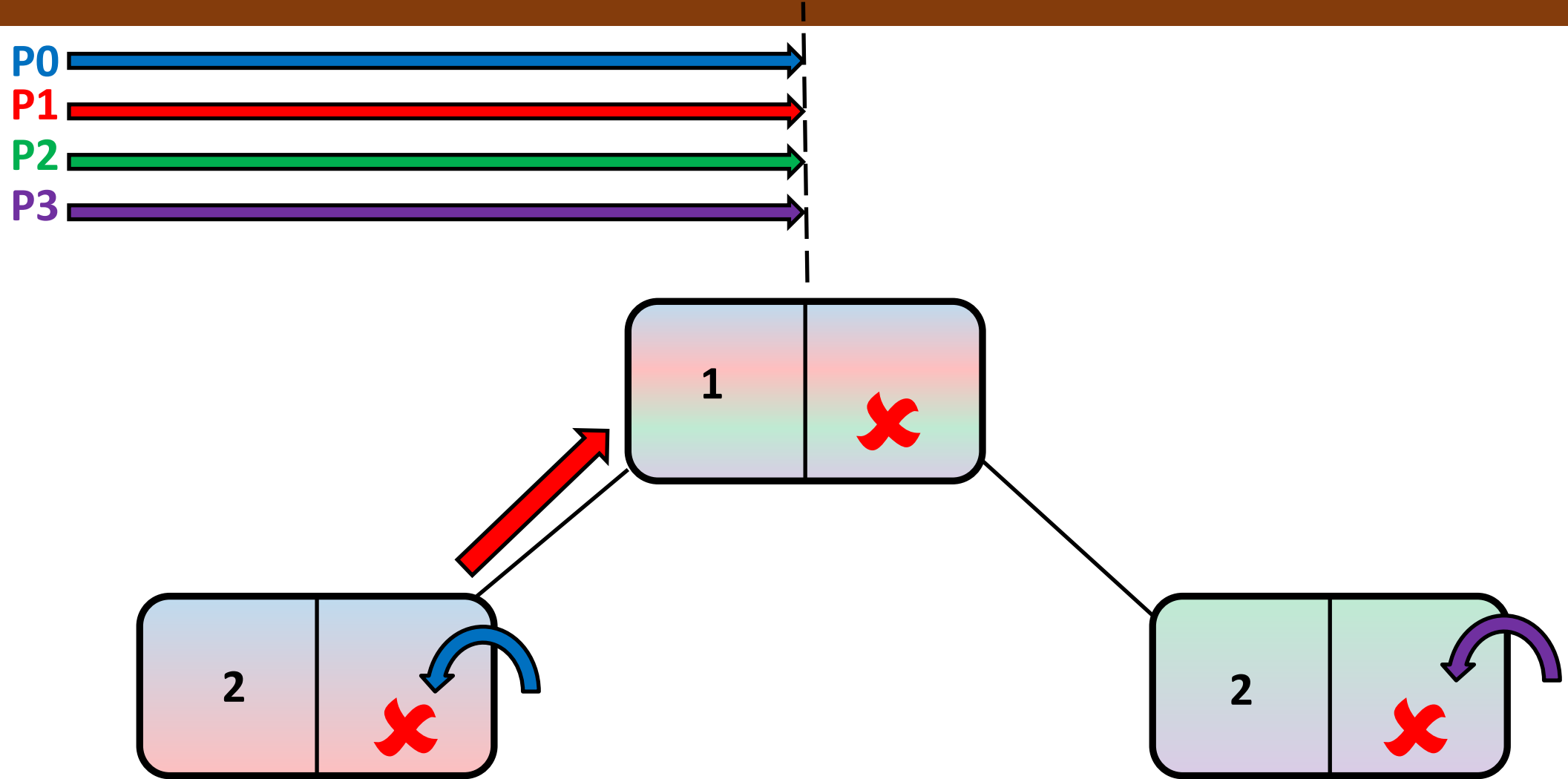
Software Combining Tree Barrier



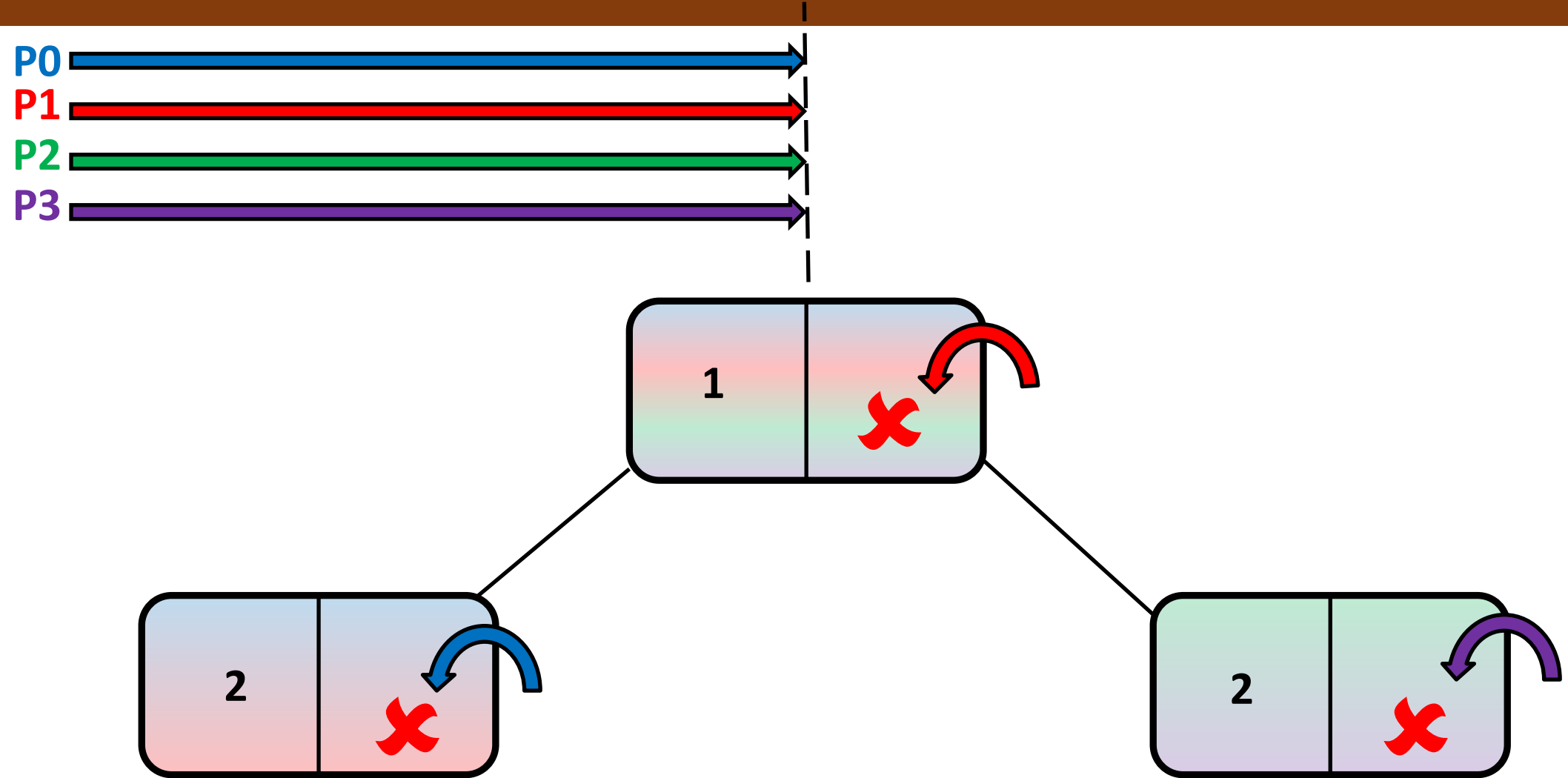
Software Combining Tree Barrier



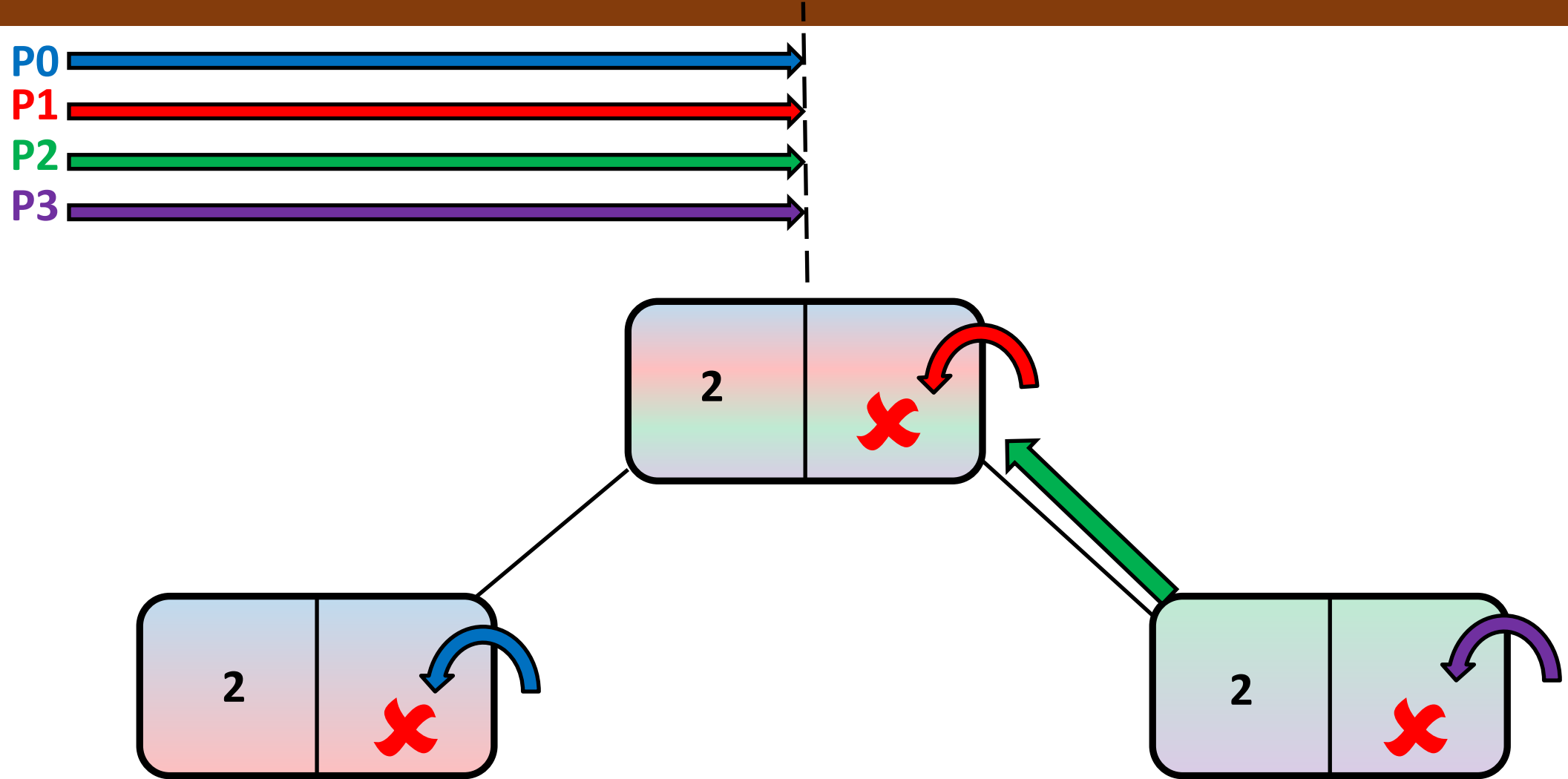
Software Combining Tree Barrier



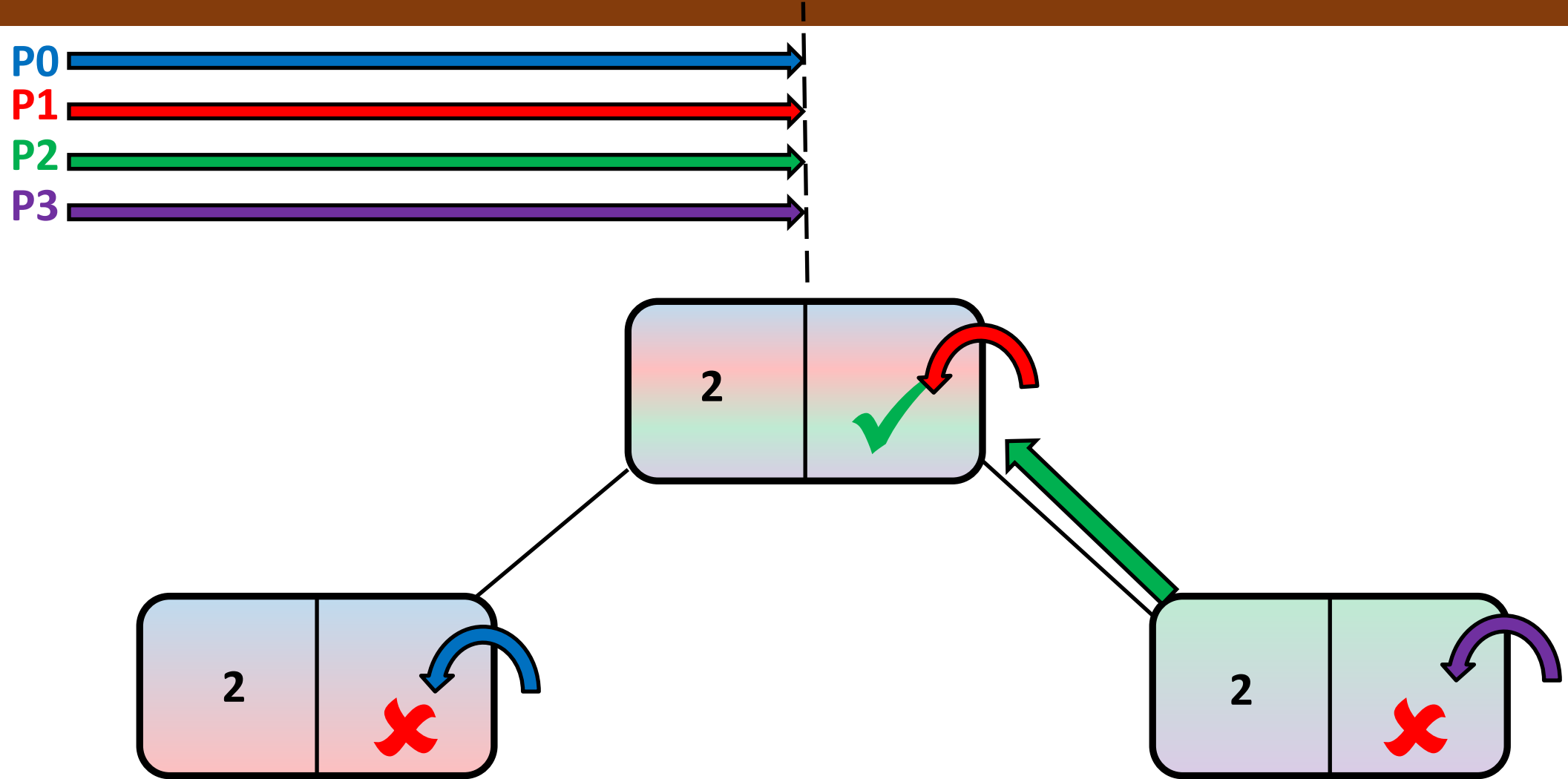
Software Combining Tree Barrier



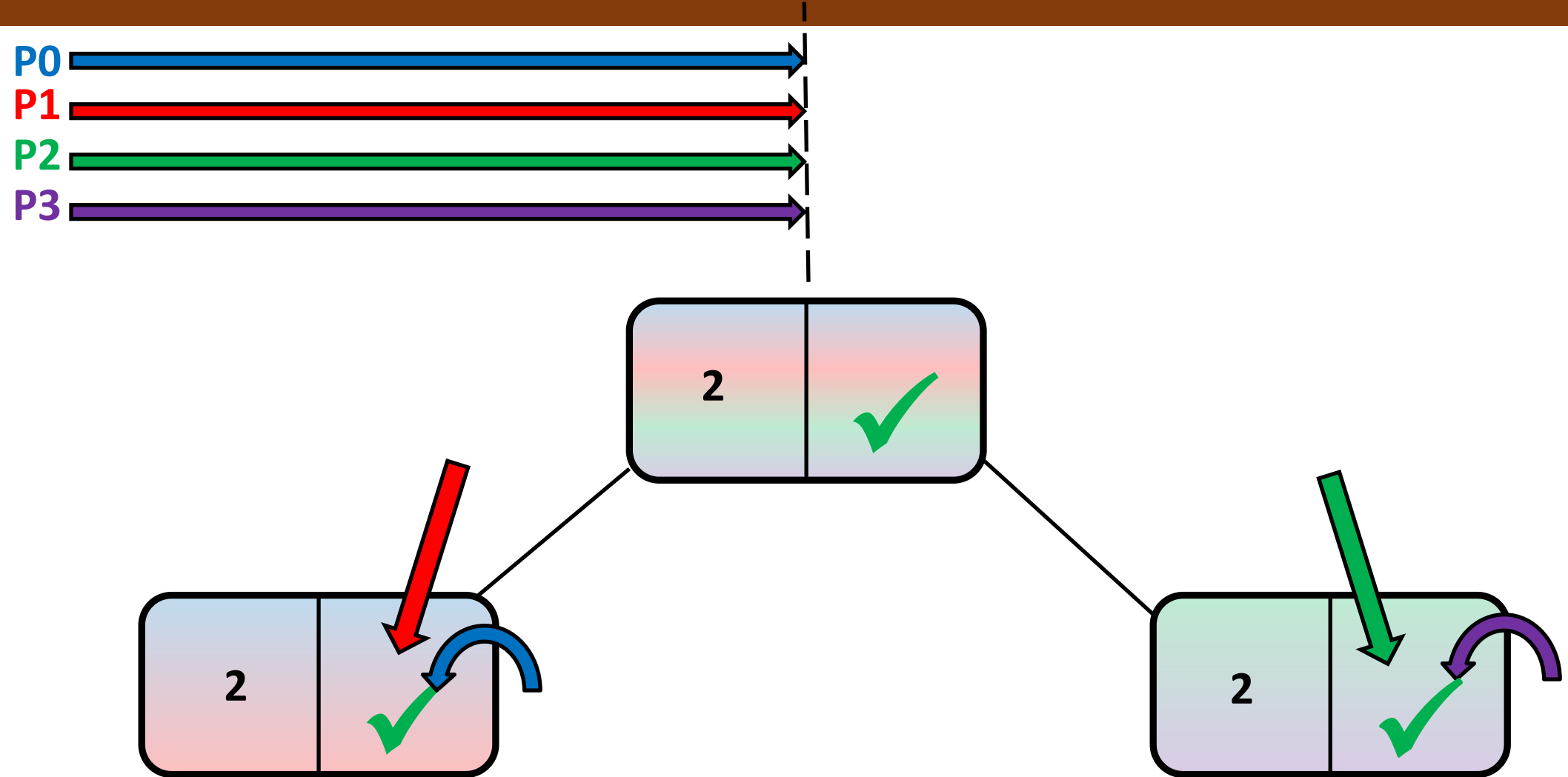
Software Combining Tree Barrier



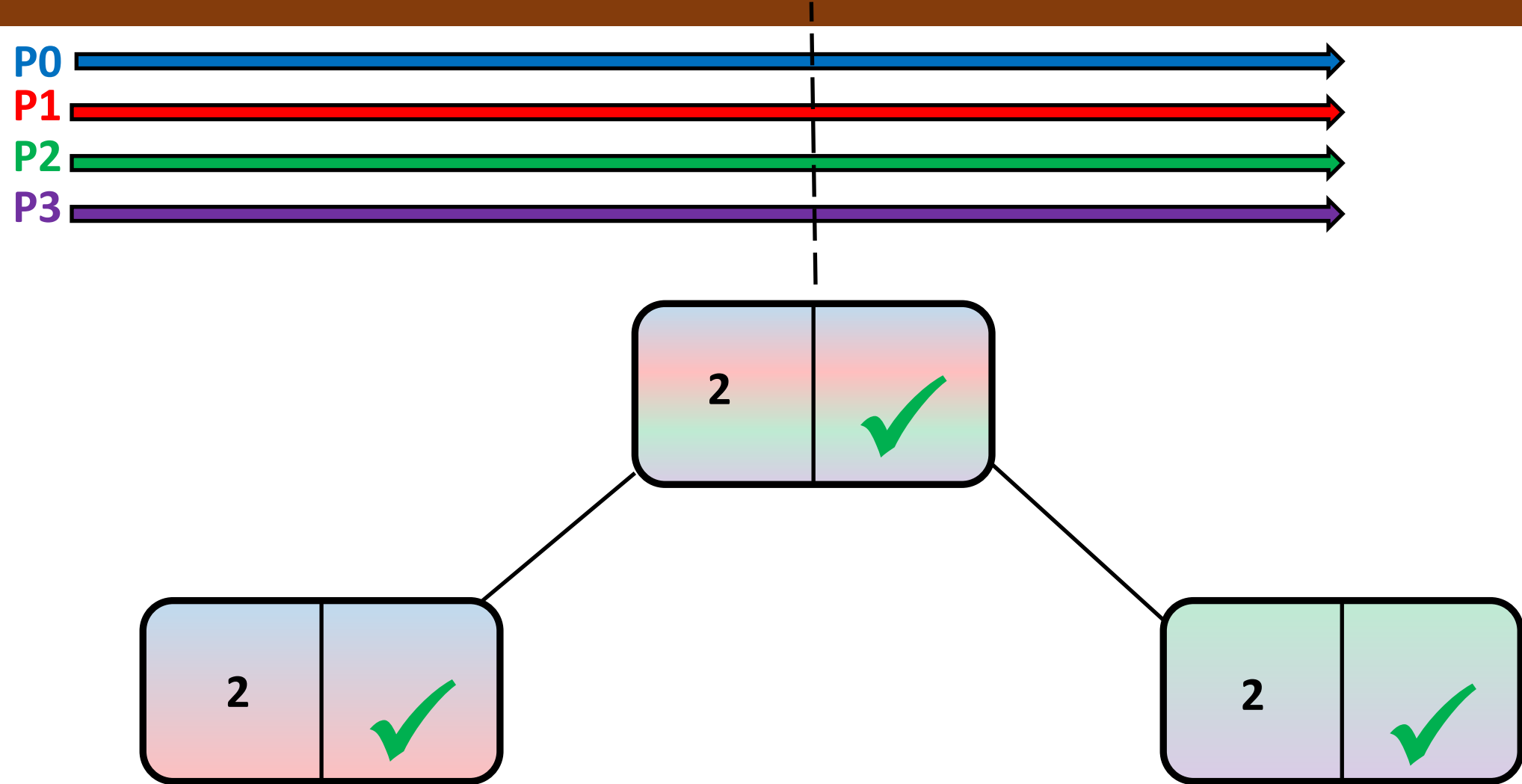
Software Combining Tree Barrier



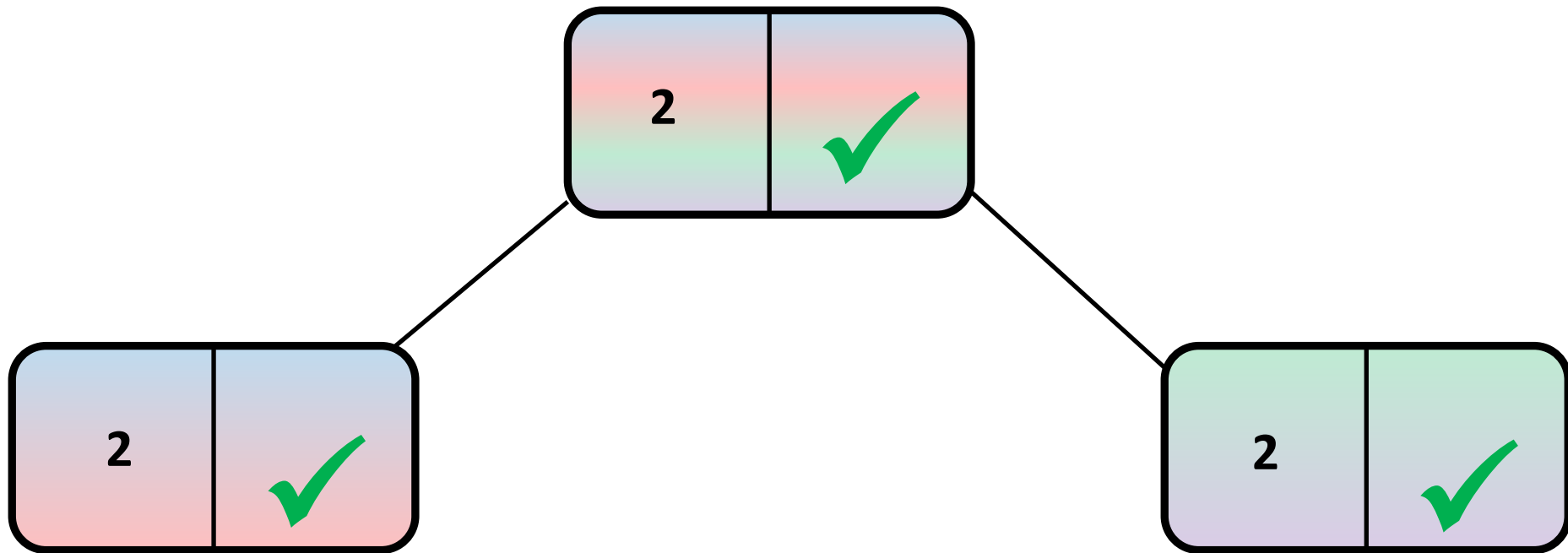
Software Combining Tree Barrier



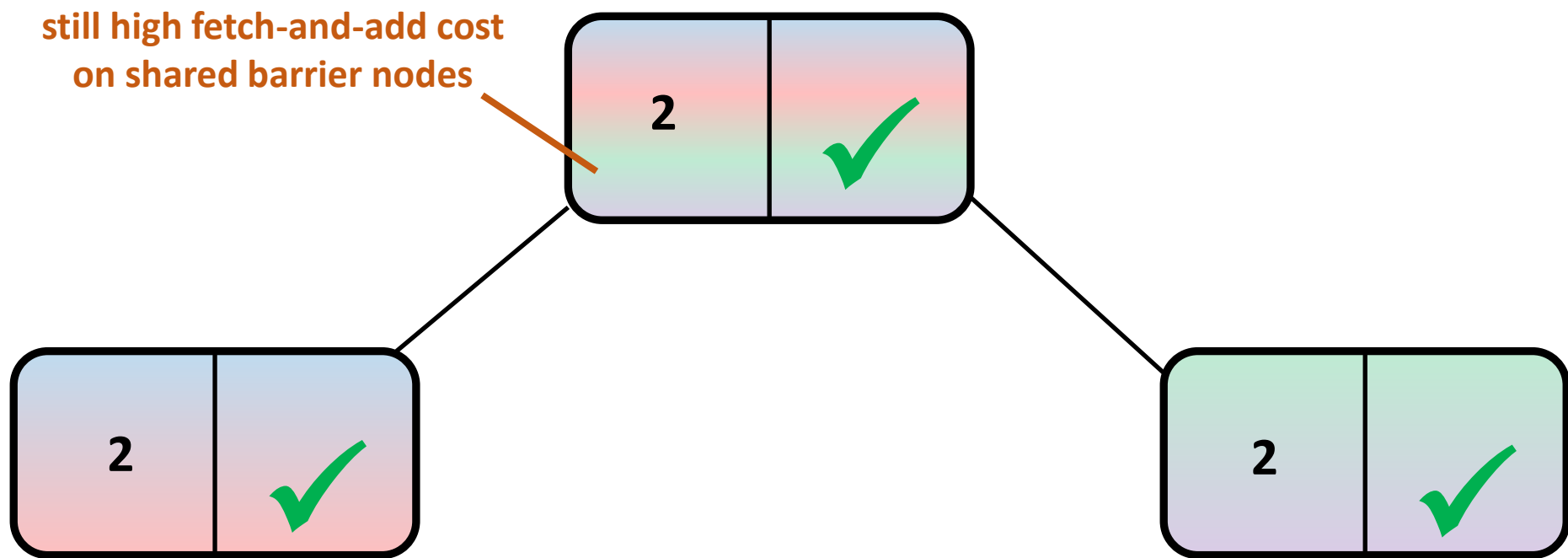
Software Combining Tree Barrier



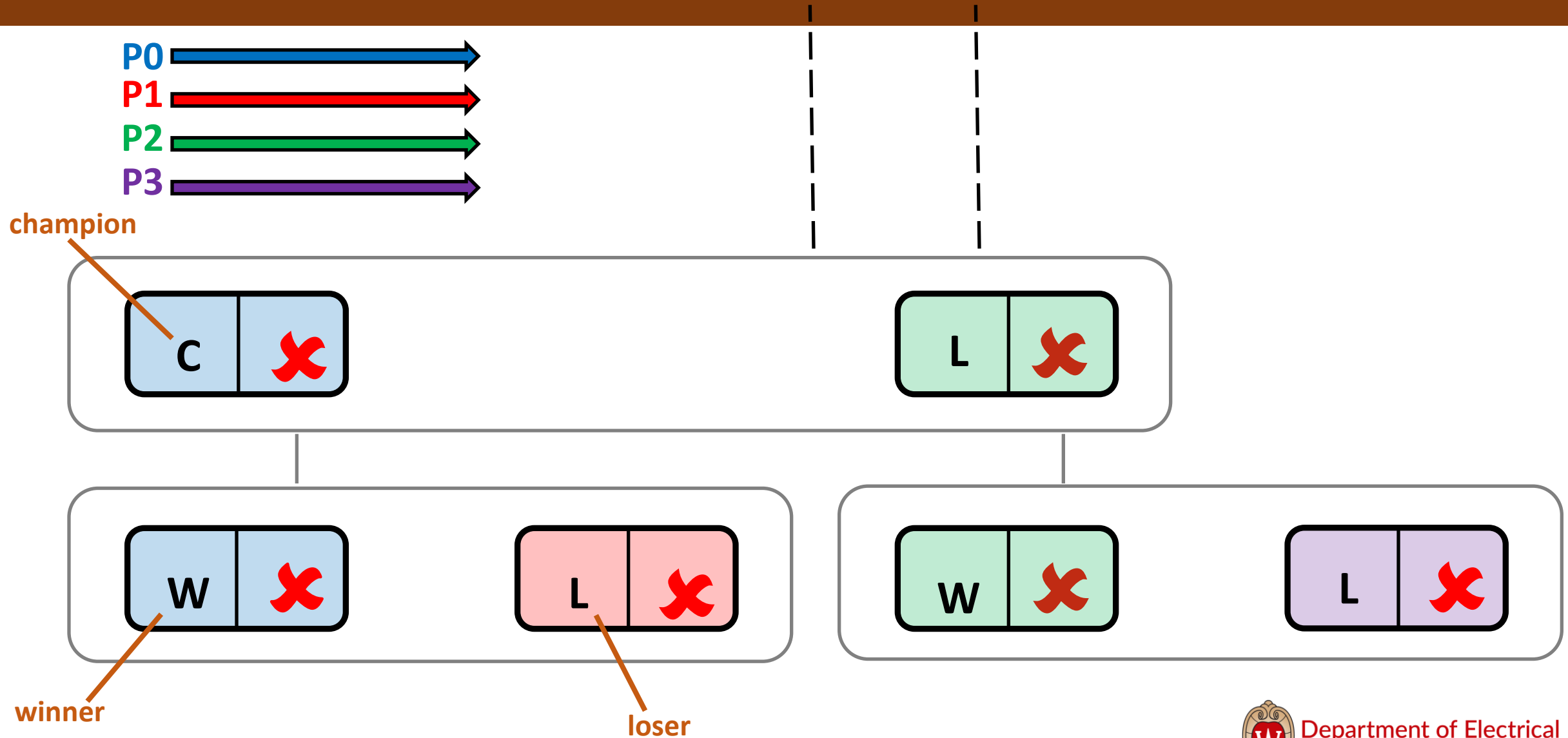
Software Combining Tree Barrier



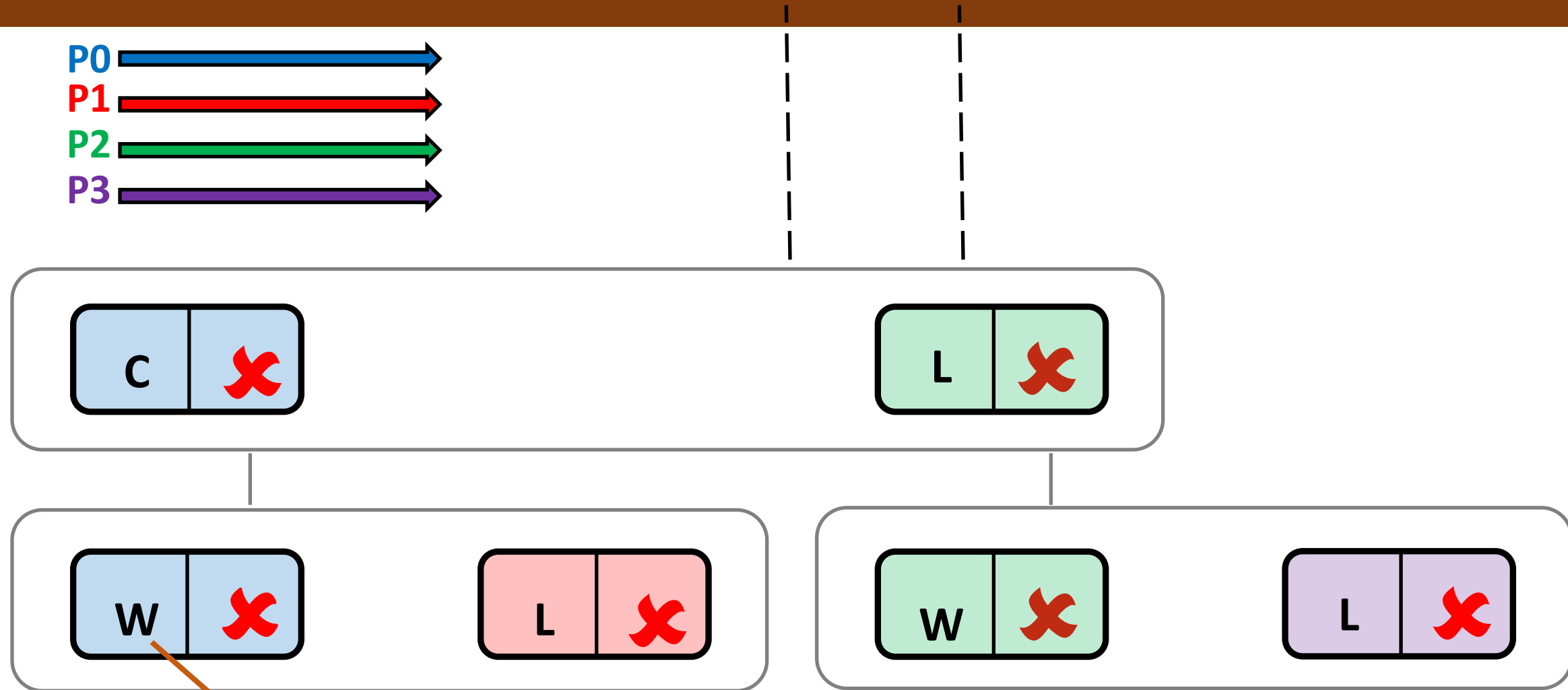
Software Combining Tree Barrier



Tournament Barrier

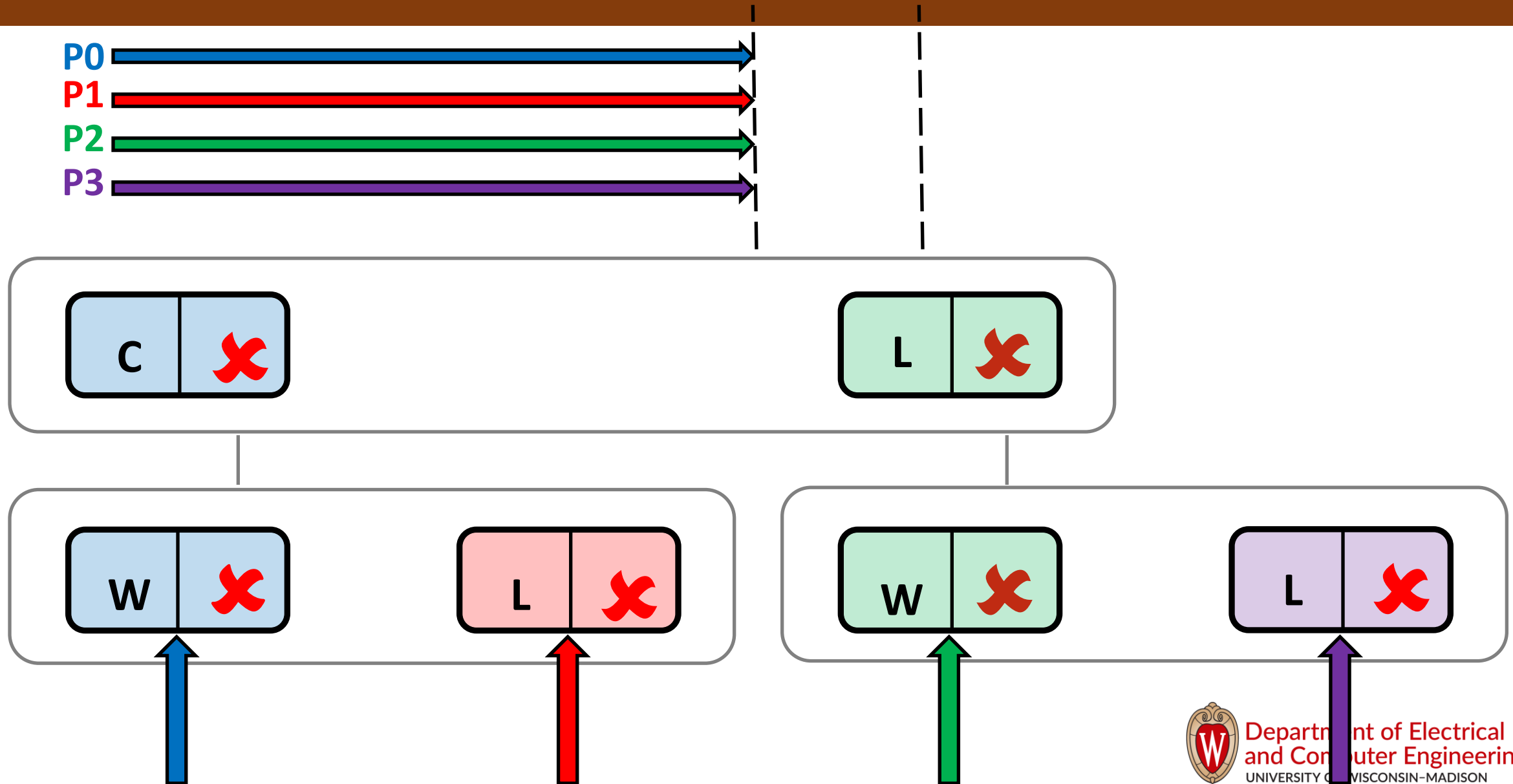


Tournament Barrier

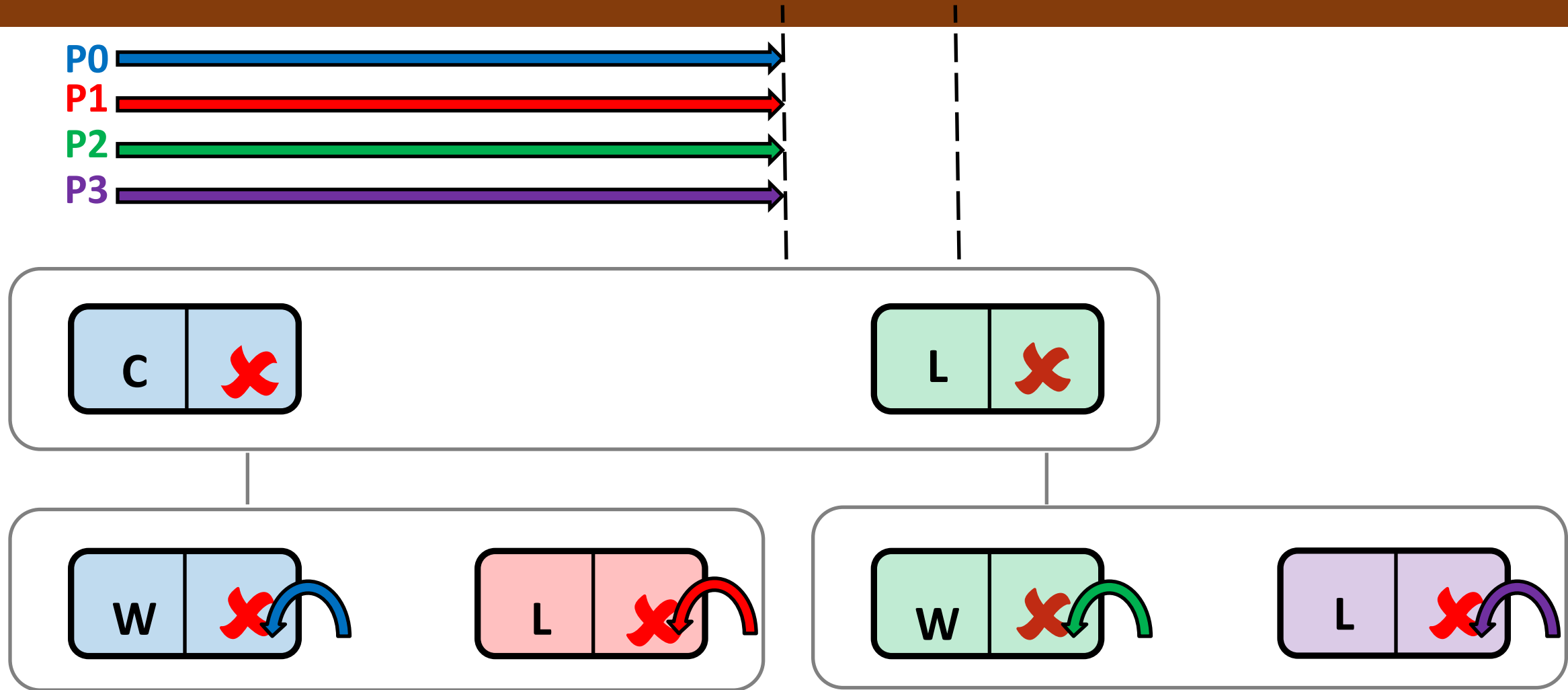


eliminate fetch-and-add by
predefining who moves up

Tournament Barrier

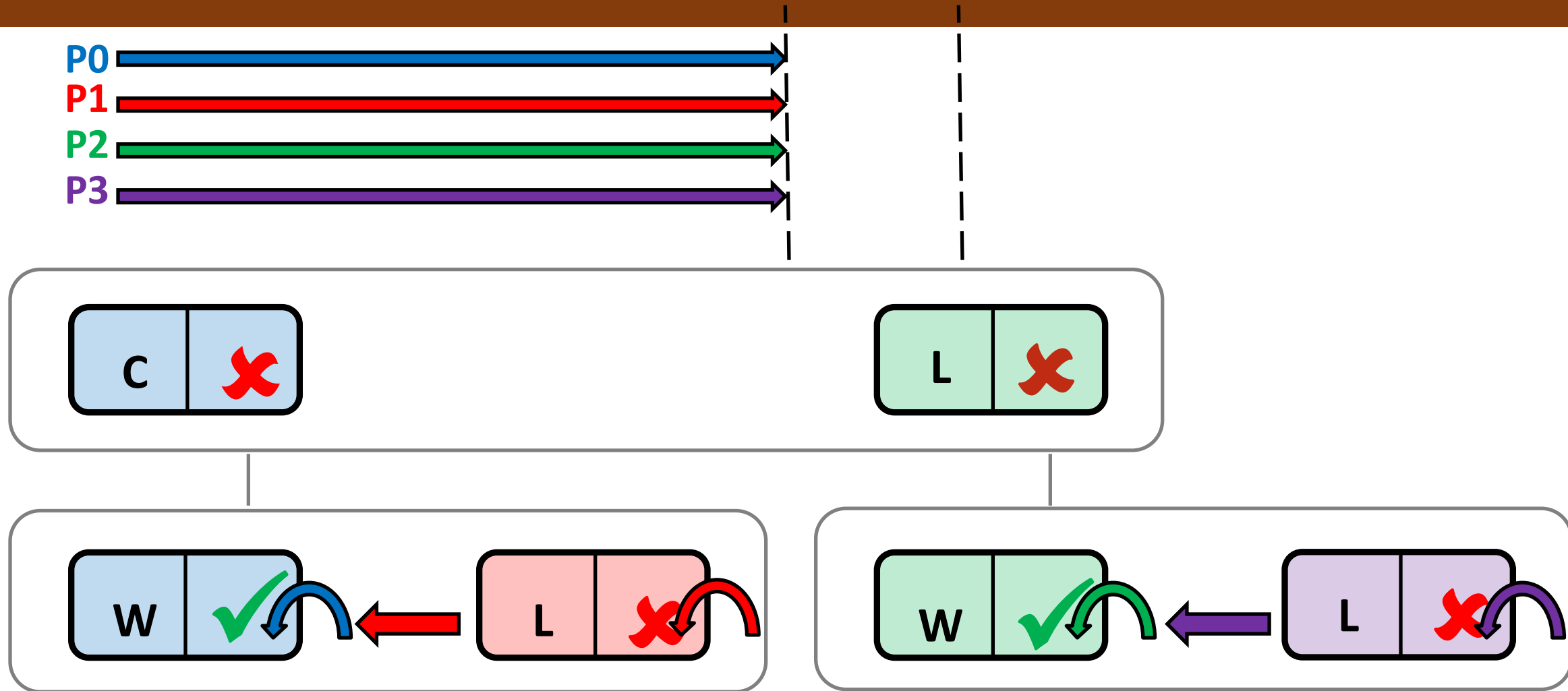


Tournament Barrier

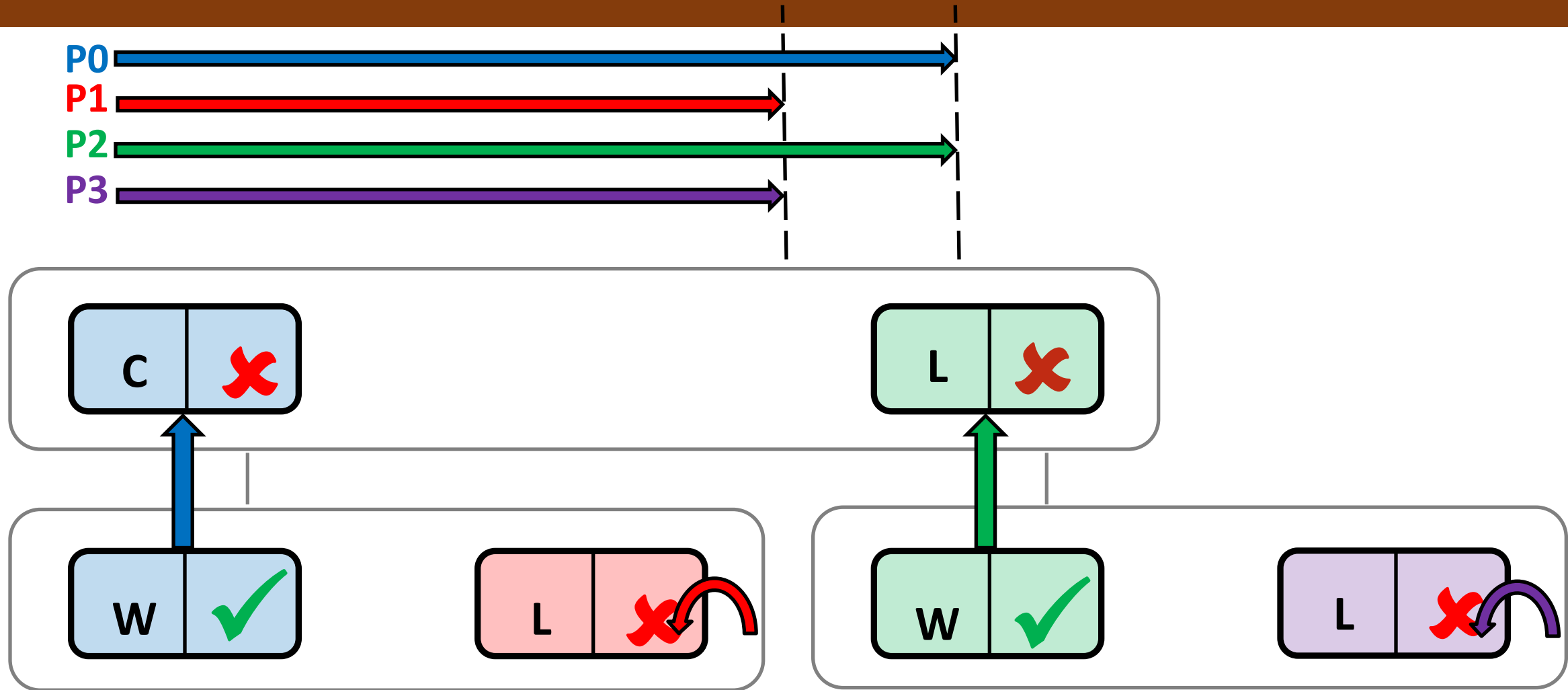


spin on private memory locations!

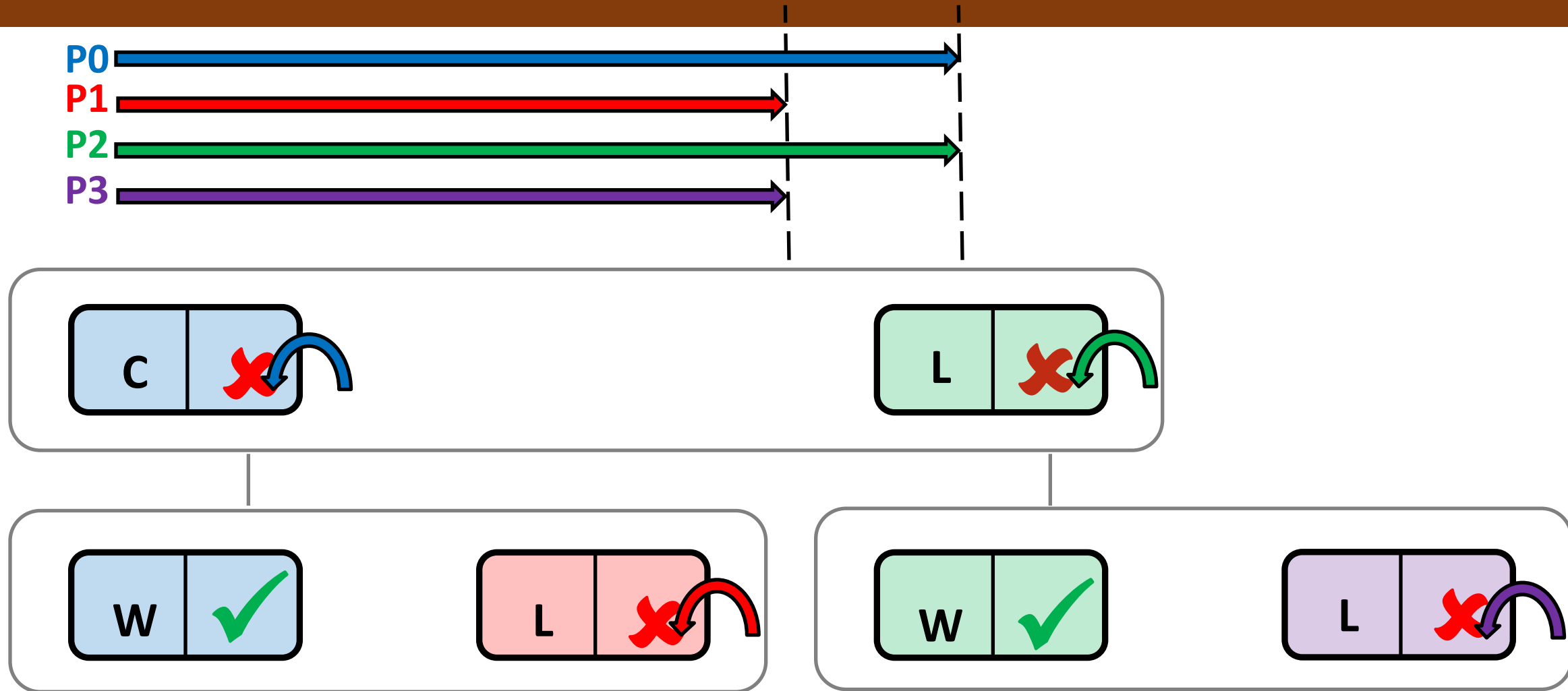
Tournament Barrier



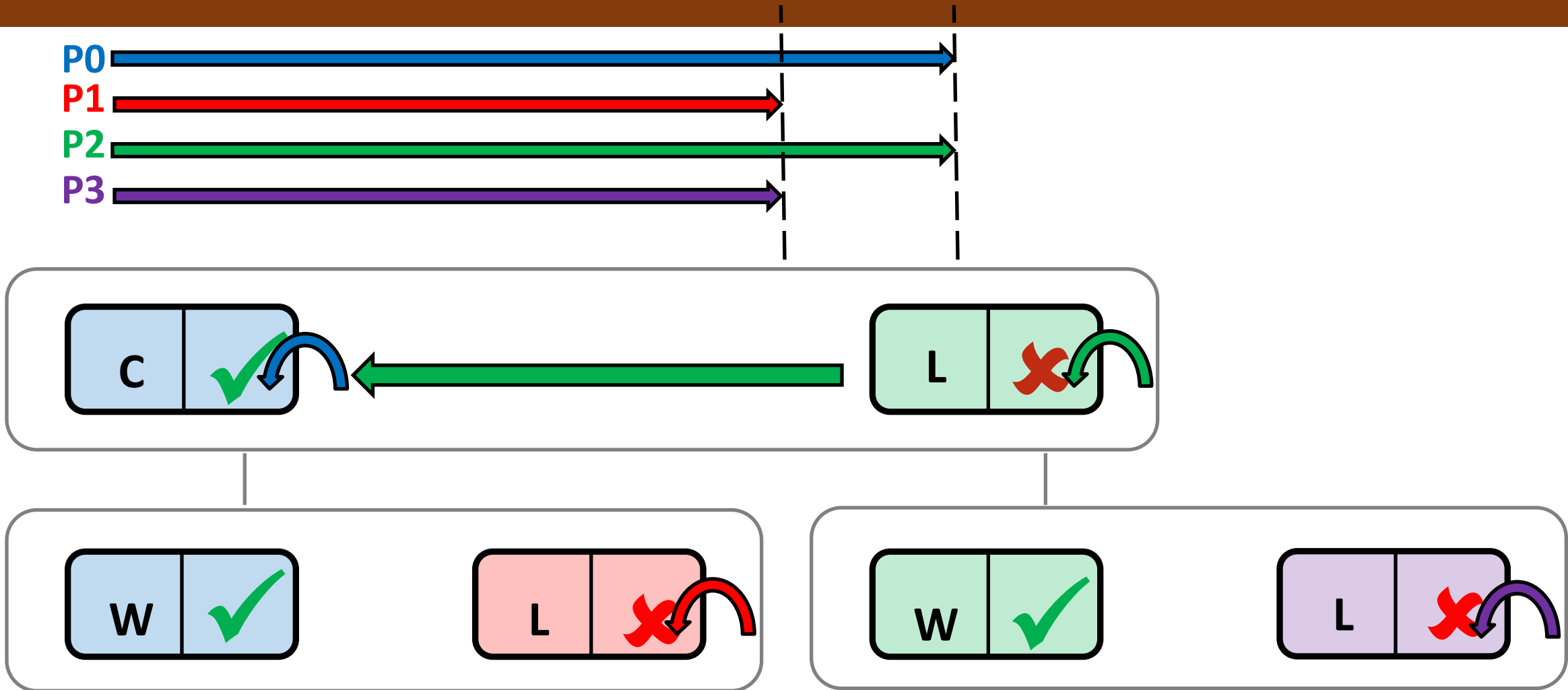
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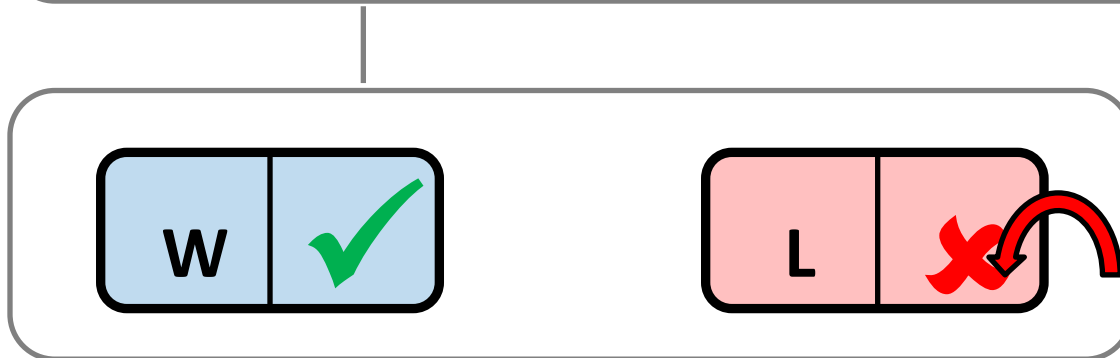
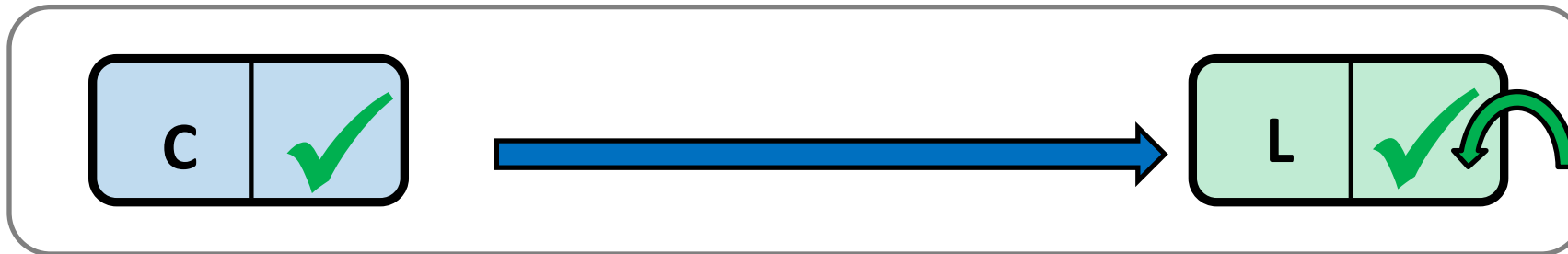
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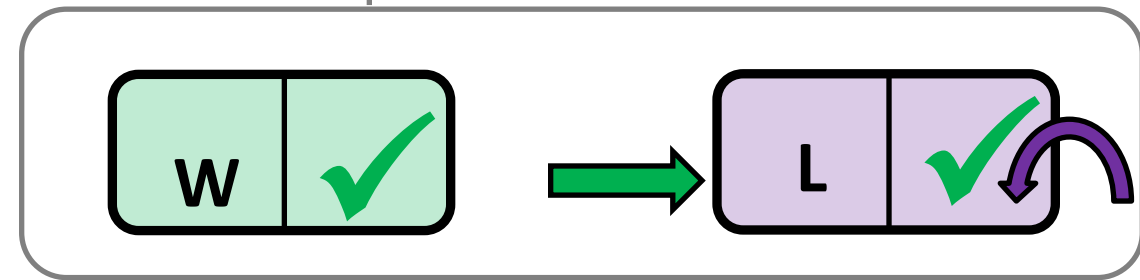
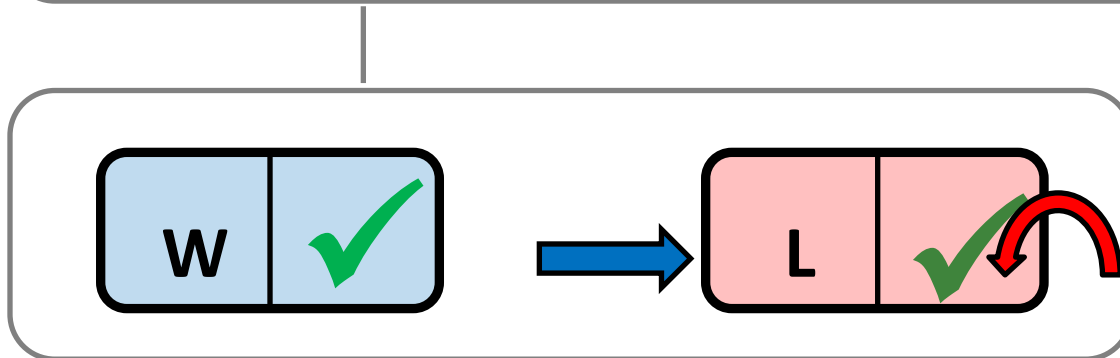
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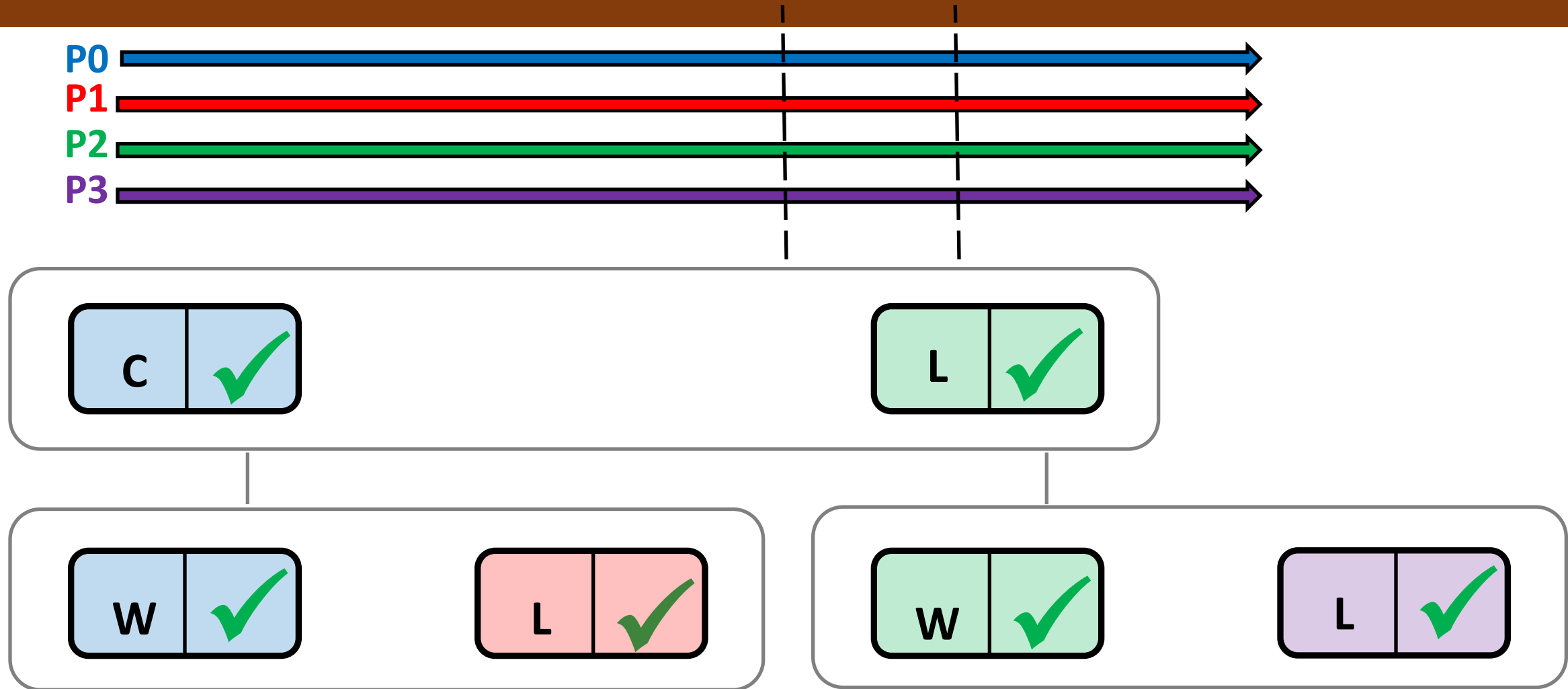
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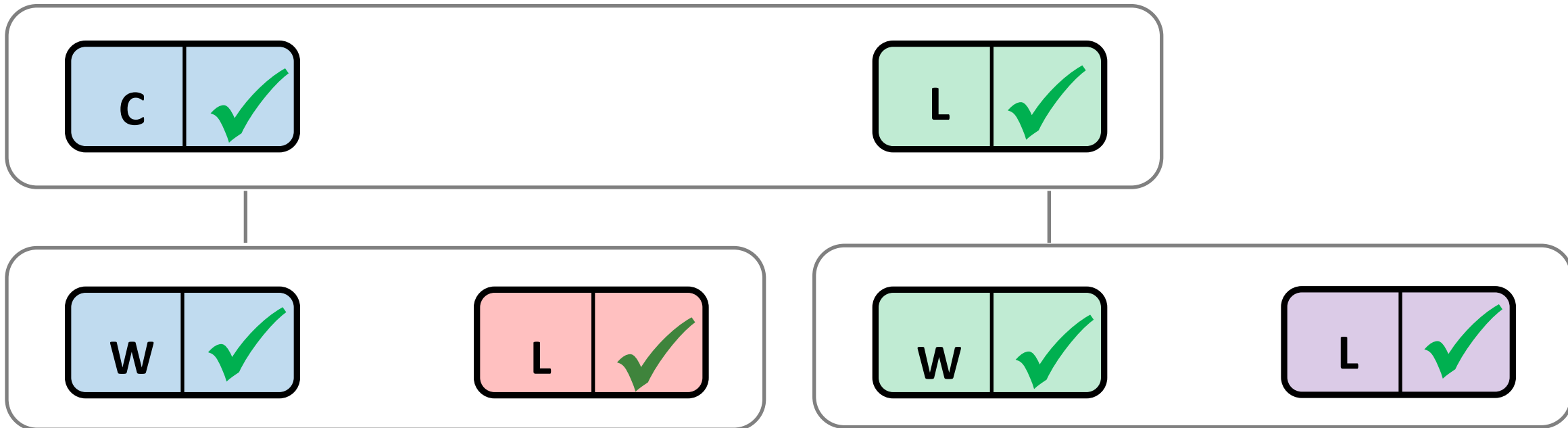
Tournament Barrier



Tournament Barrier

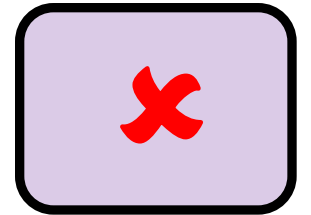
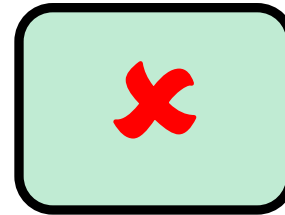
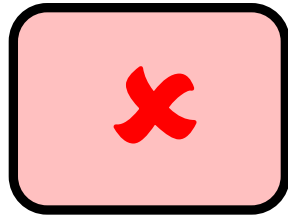
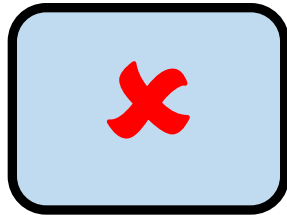
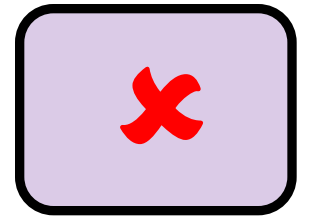
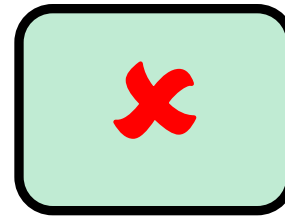
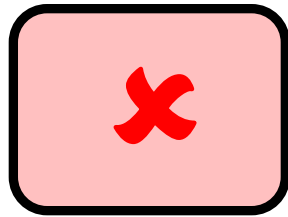
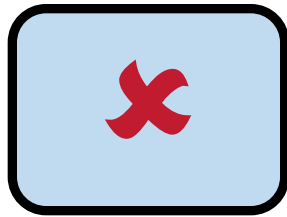


Tournament Barrier



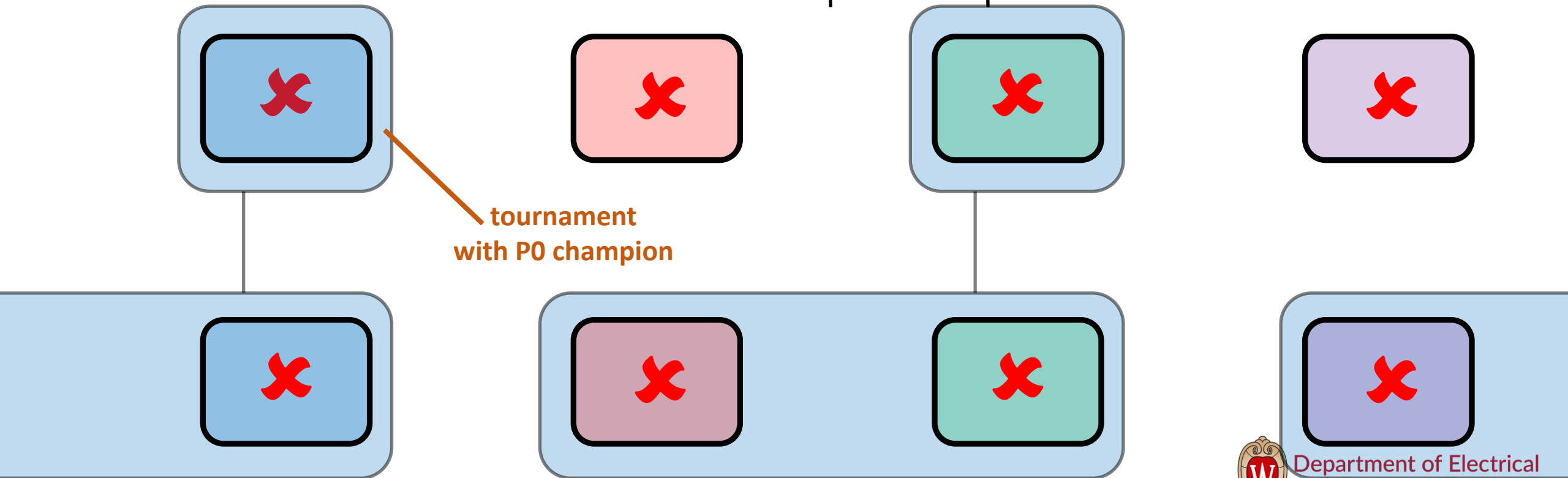
Dissemination Barrier

P0 →
P1 →
P2 →
P3 →



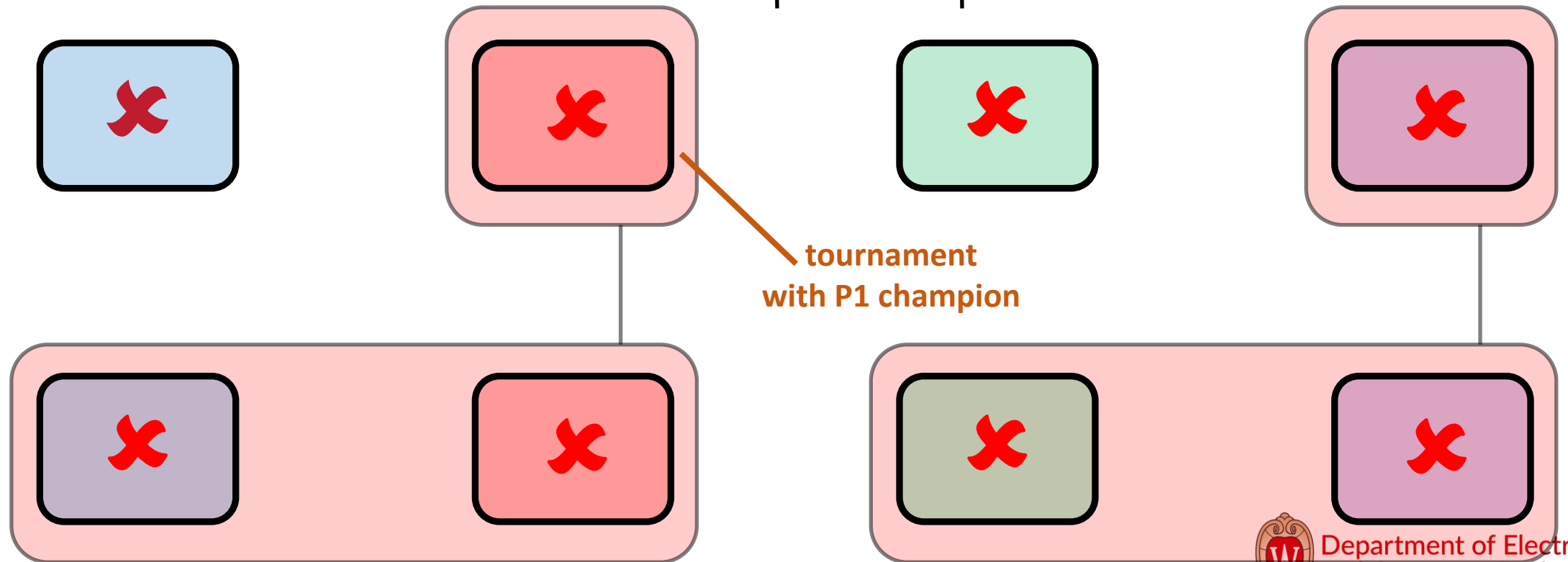
Dissemination Barrier

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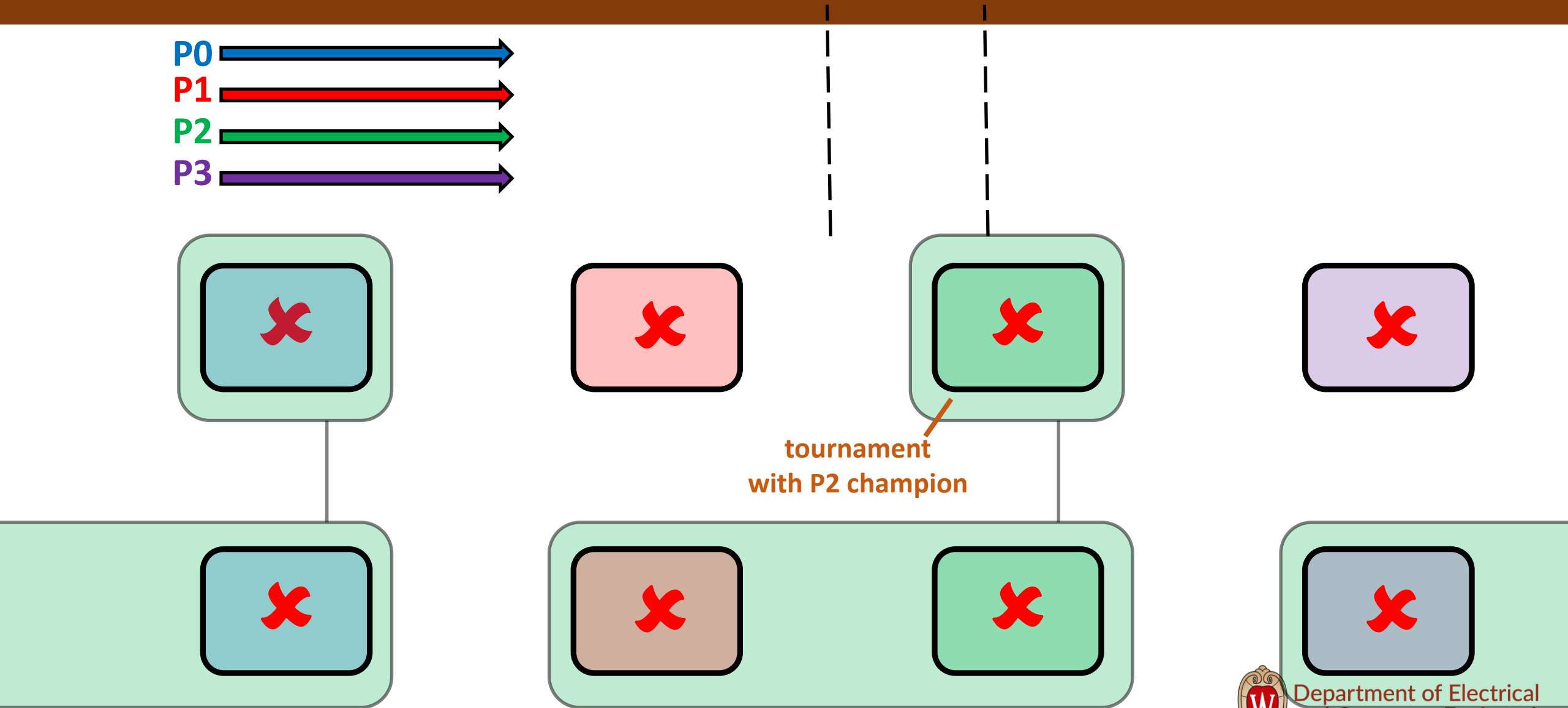
Dissemination Barrier

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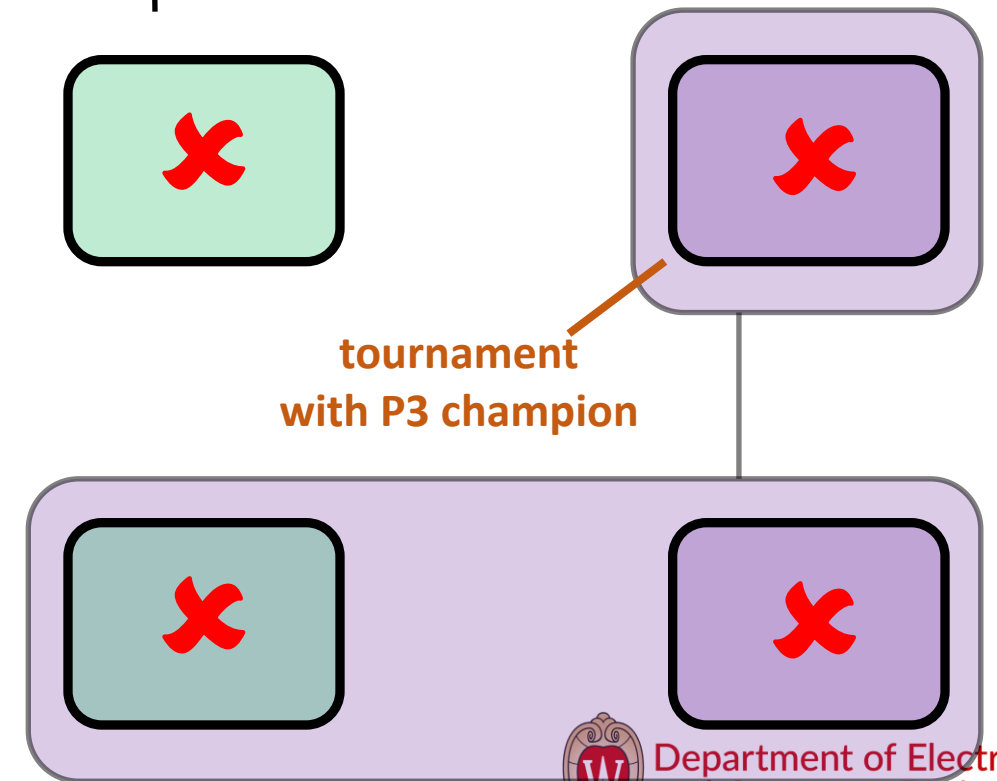
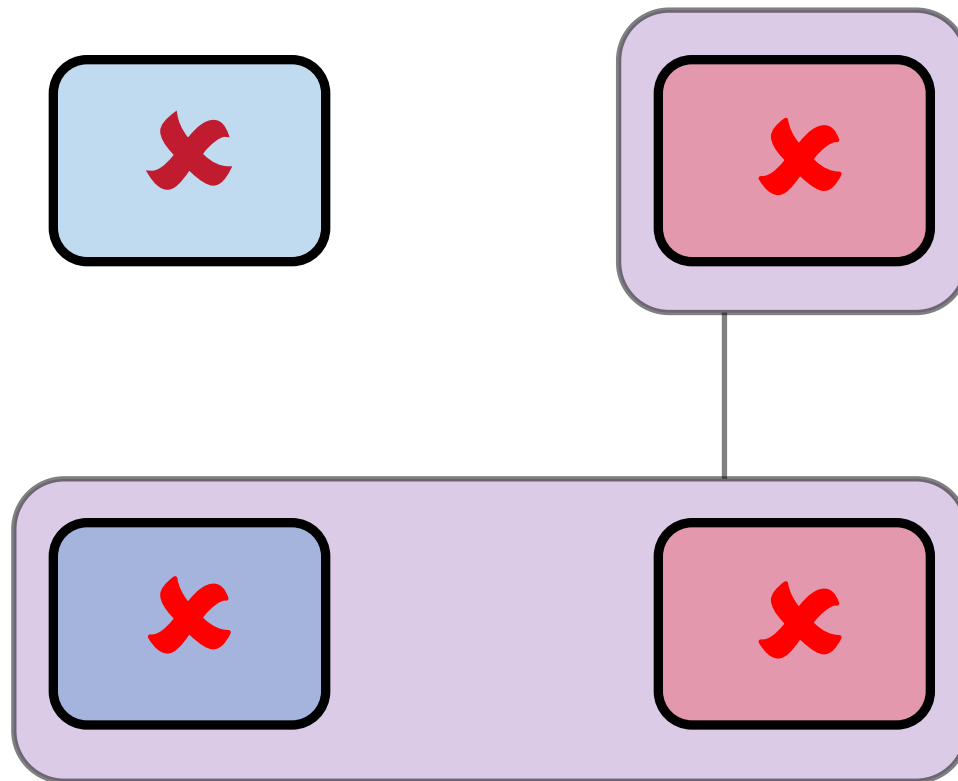
Dissemination Barrier

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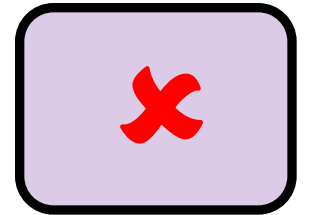
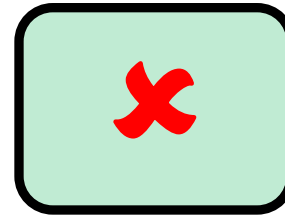
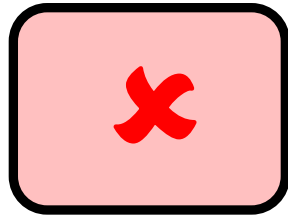
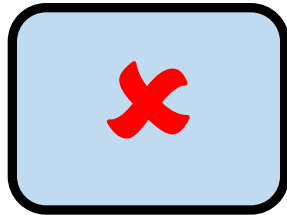
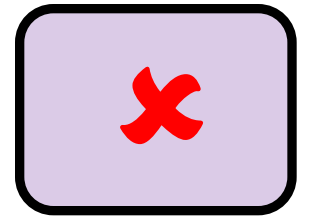
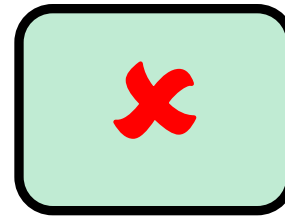
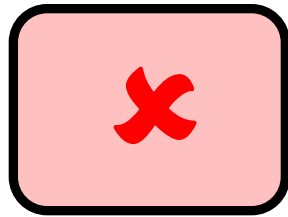
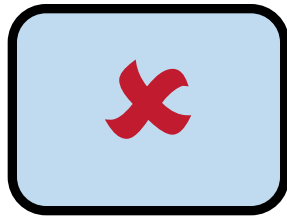
Dissemination Barrier

P0 →
P1 →
P2 →
P3 →

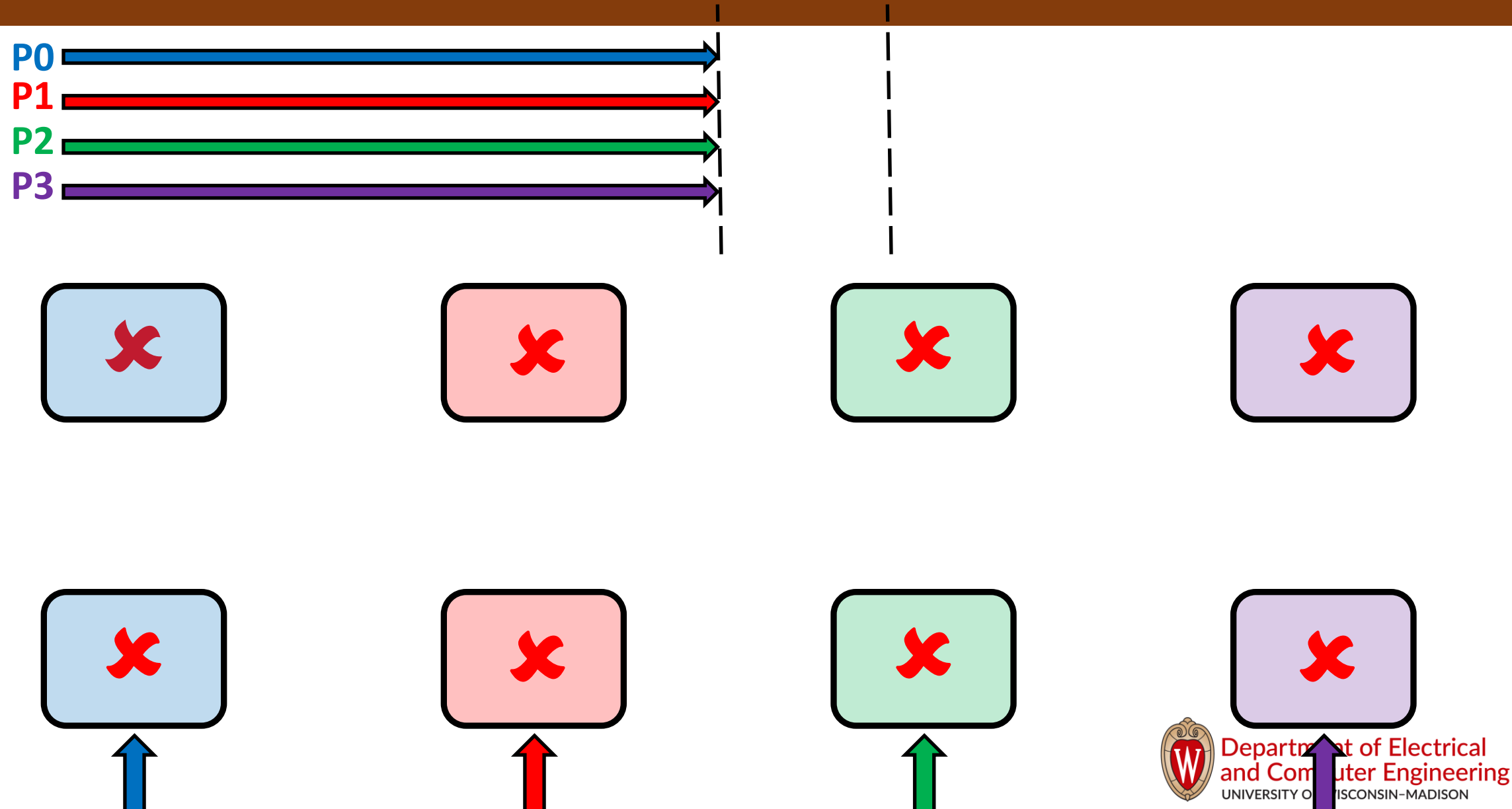


Dissemination Barrier

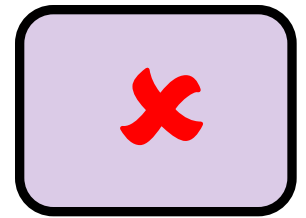
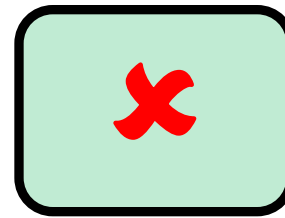
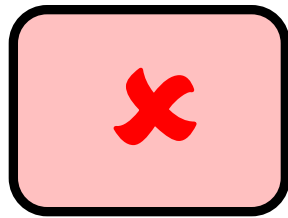
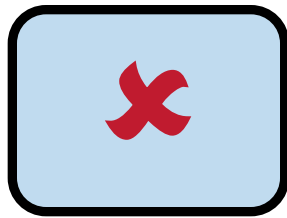
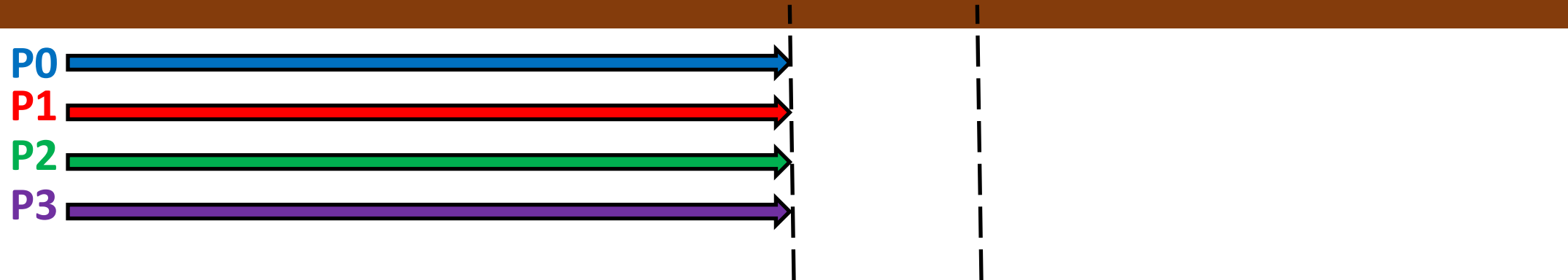
P0 →
P1 →
P2 →
P3 →



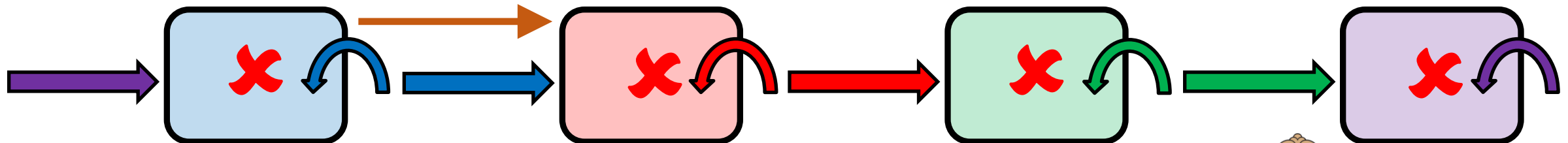
Dissemination Barrier



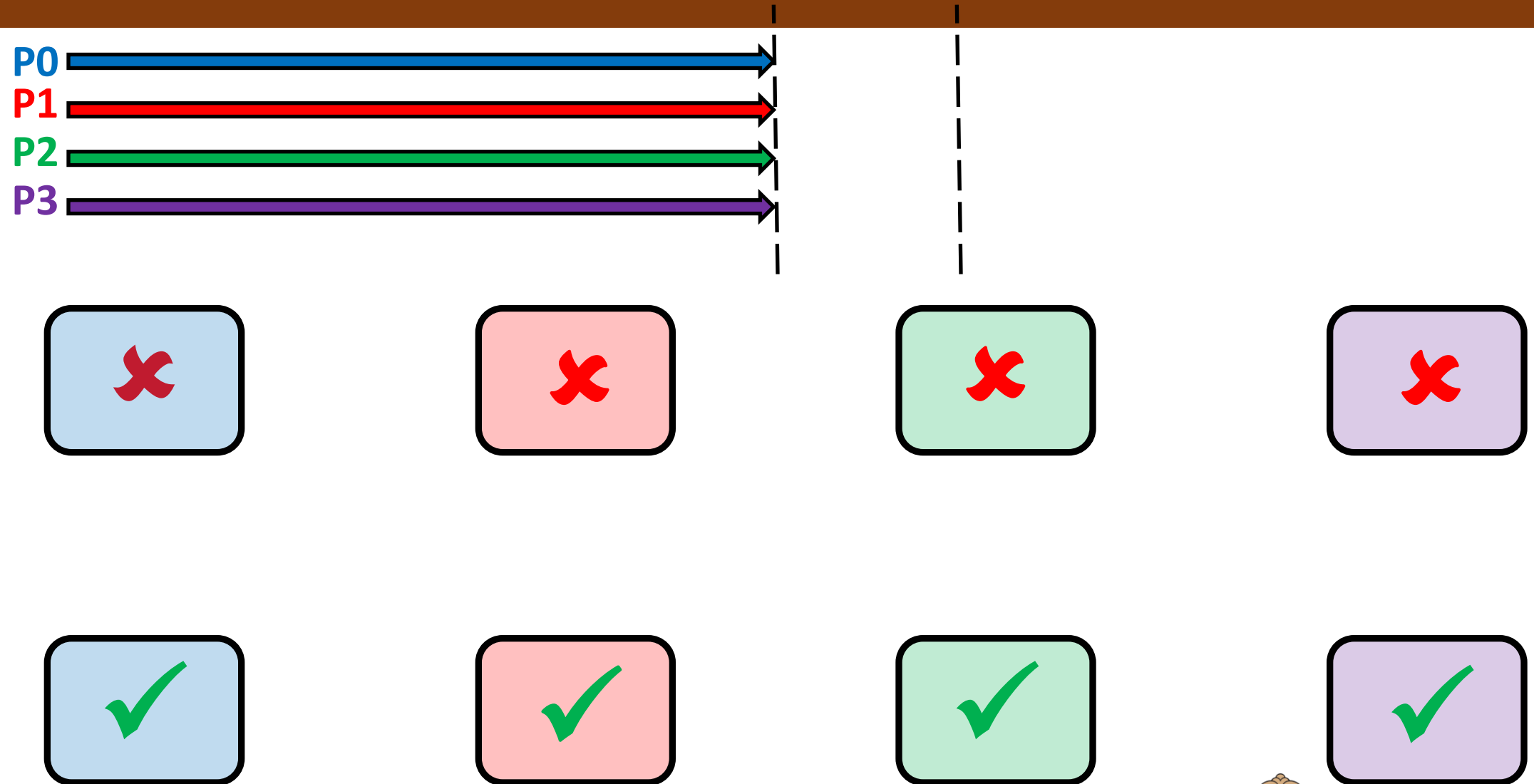
Dissemination Barrier



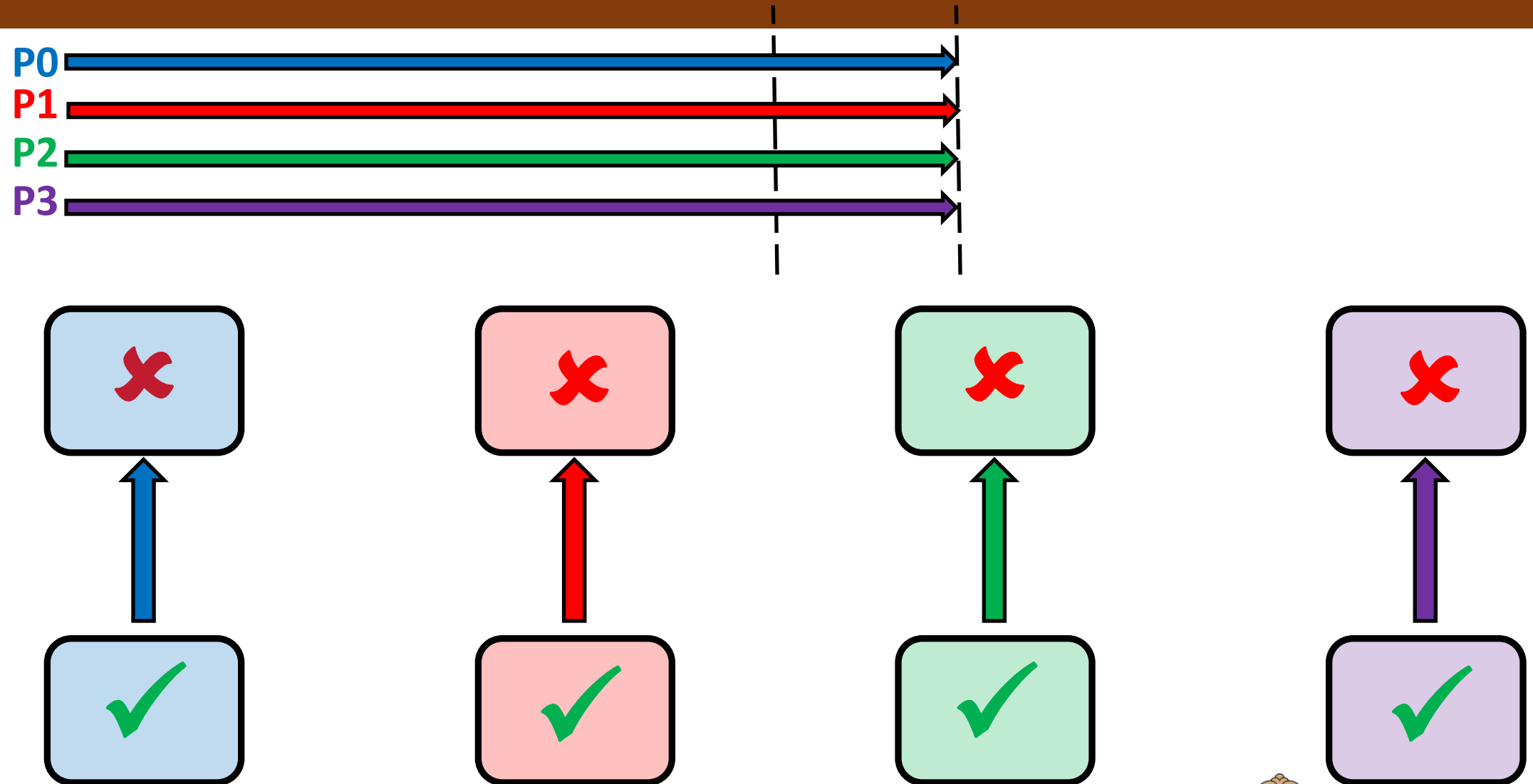
communication pattern
for loser threads



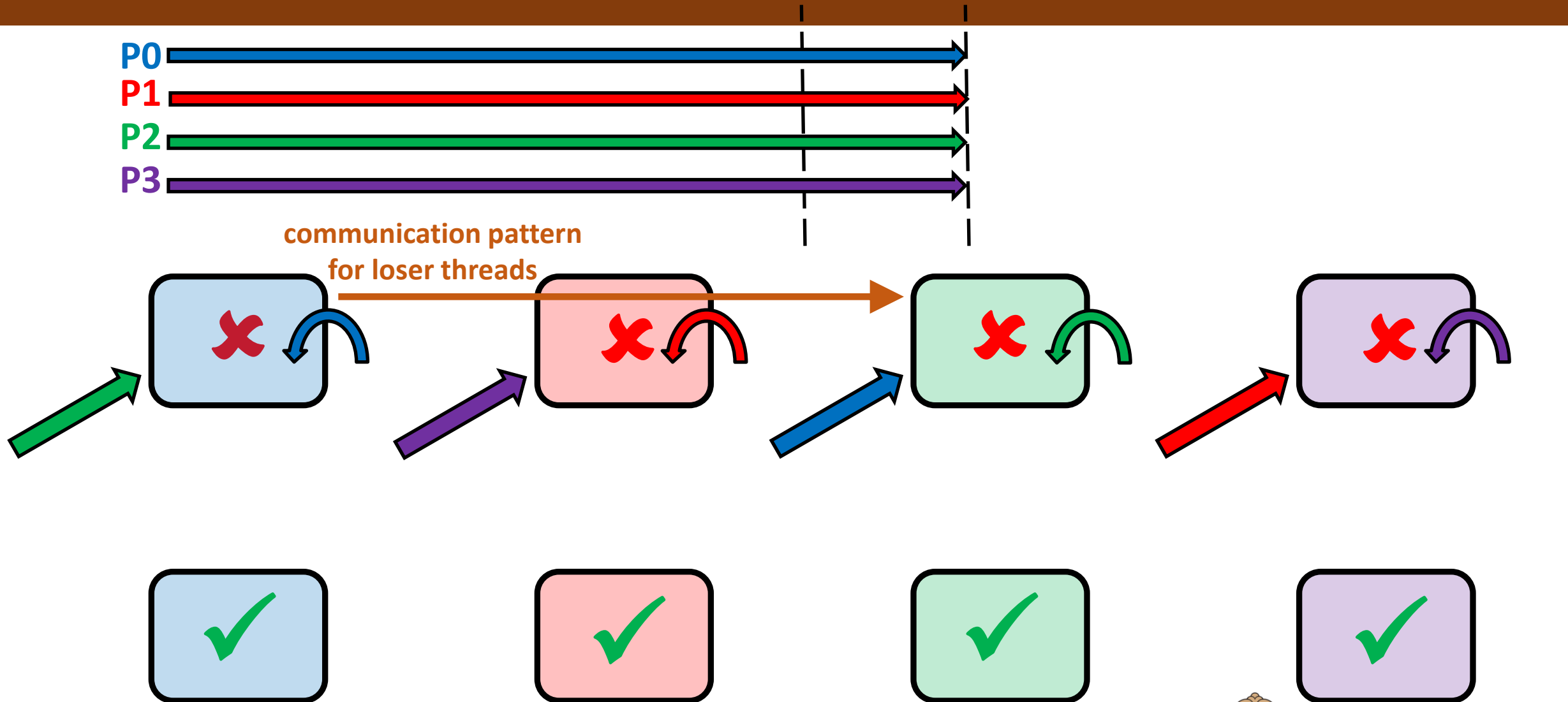
Dissemination Barrier



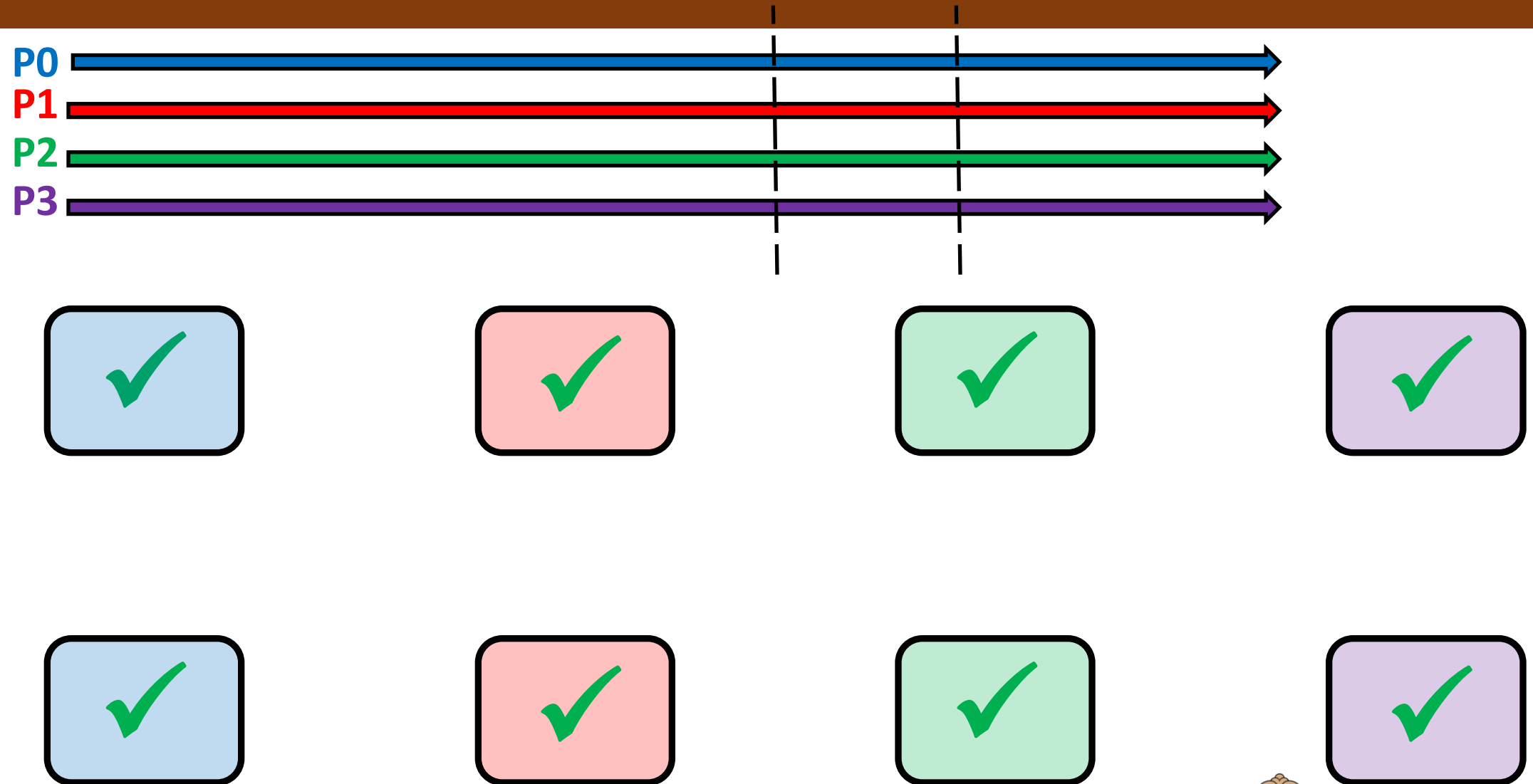
Dissemination Barrier



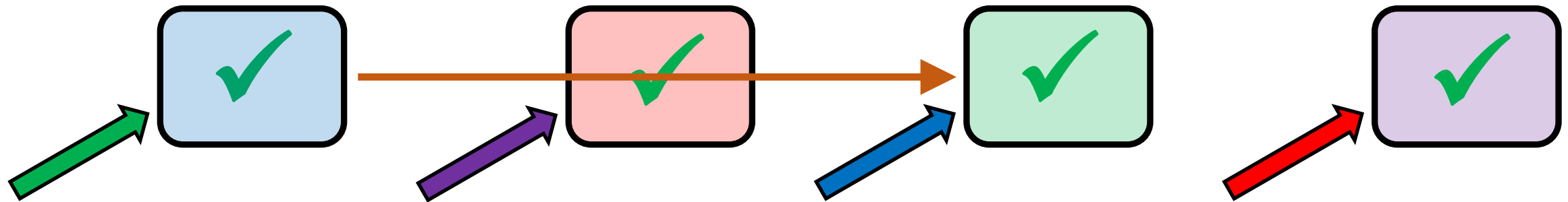
Dissemination Barrier



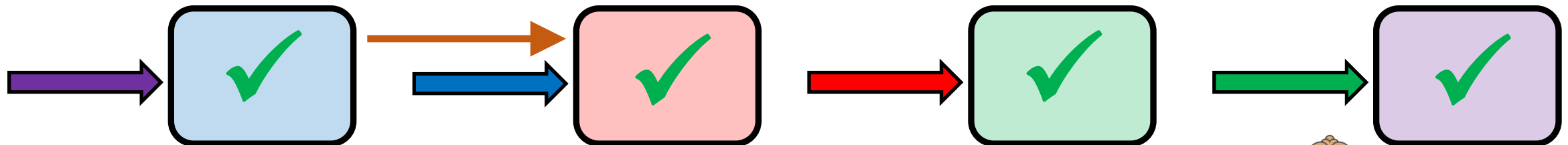
Dissemination Barrier



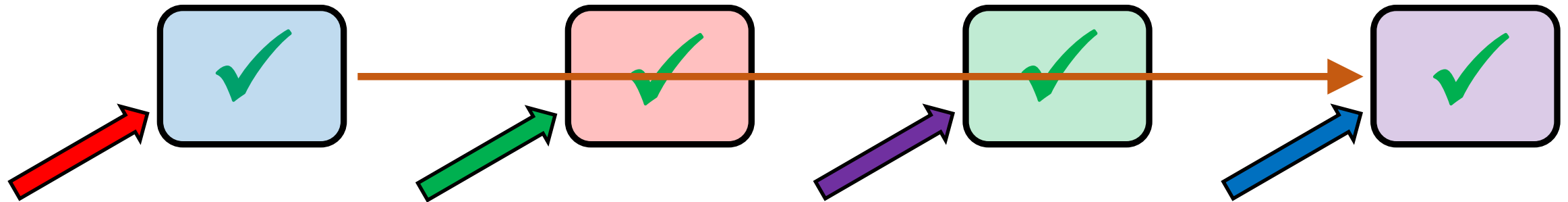
Dissemination Barrier



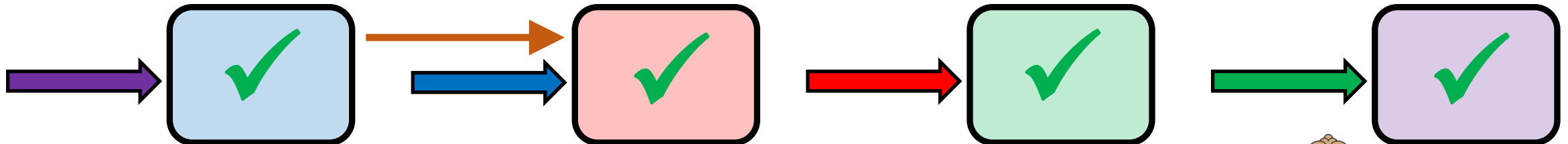
communication pattern
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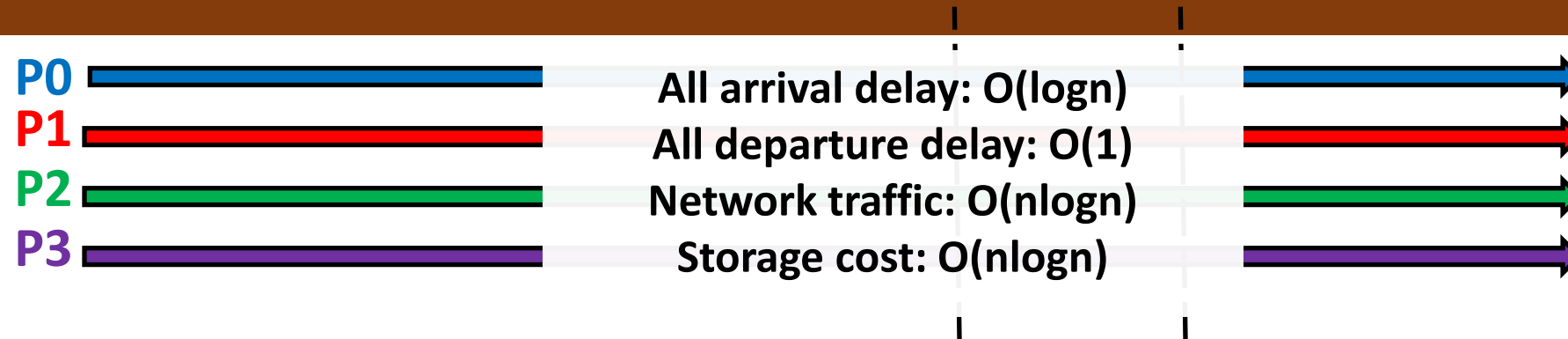
Dissemination Barrier



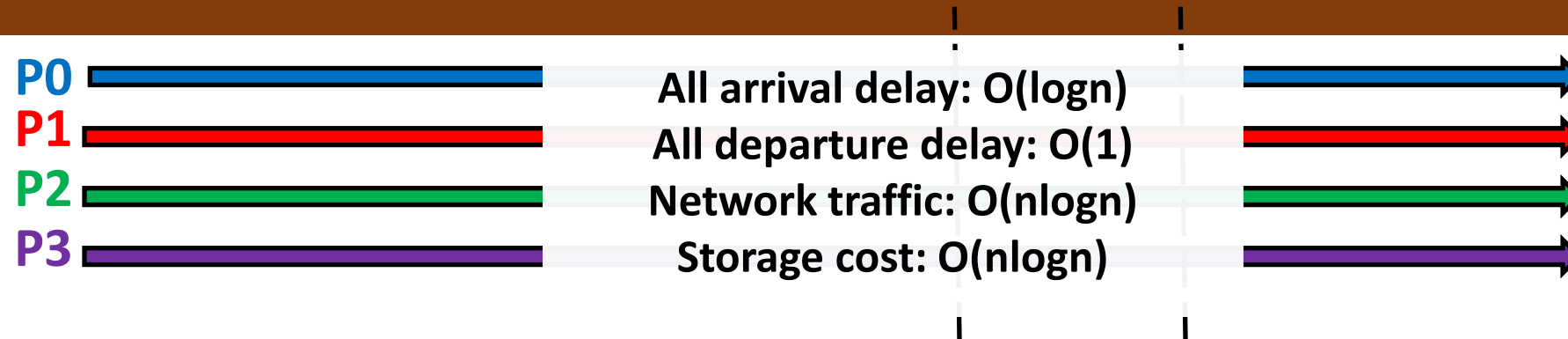
would this loser communication pattern
work instead?



Dissemination Barrier



Dissemination Barrier

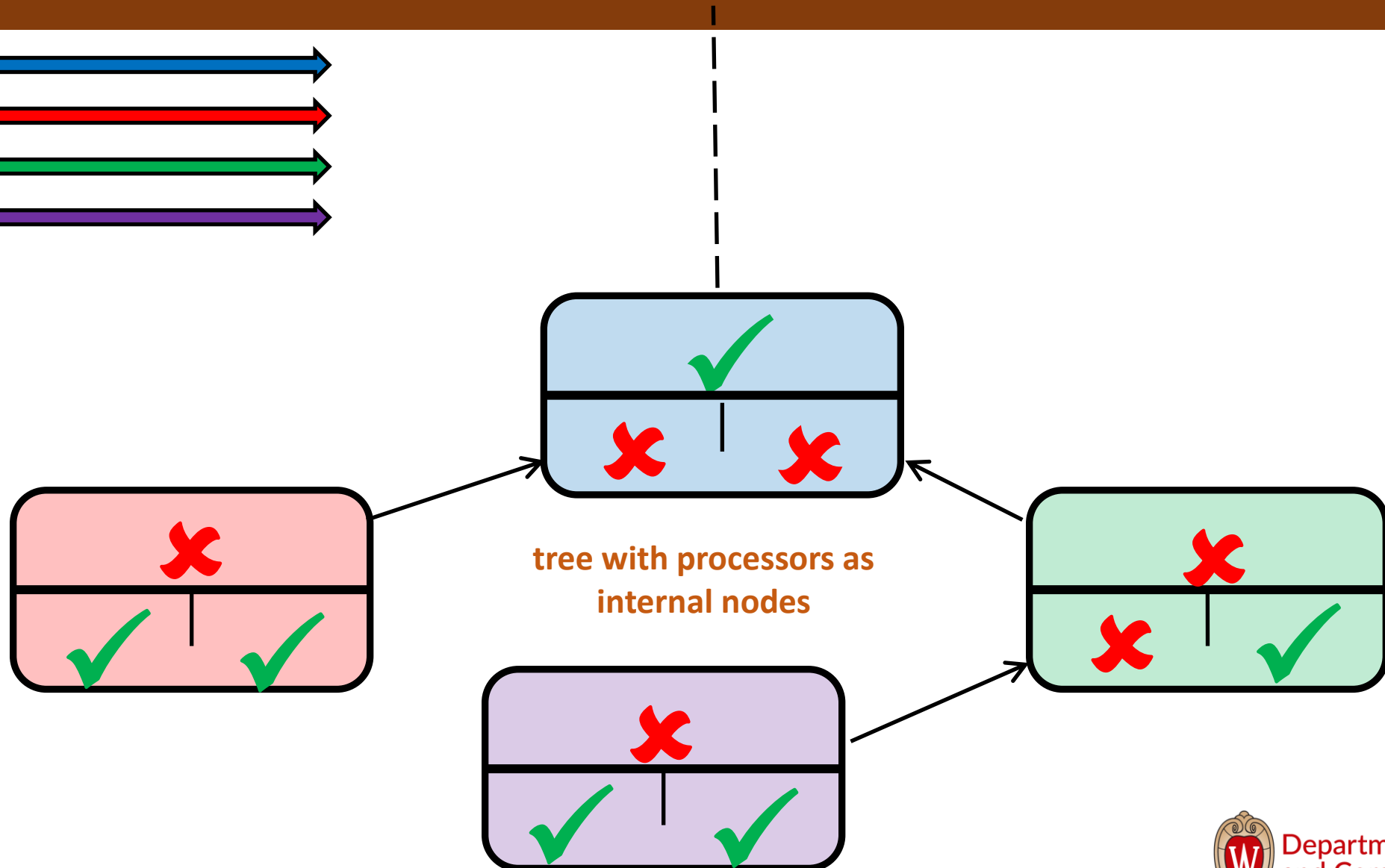


high network and storage cost!



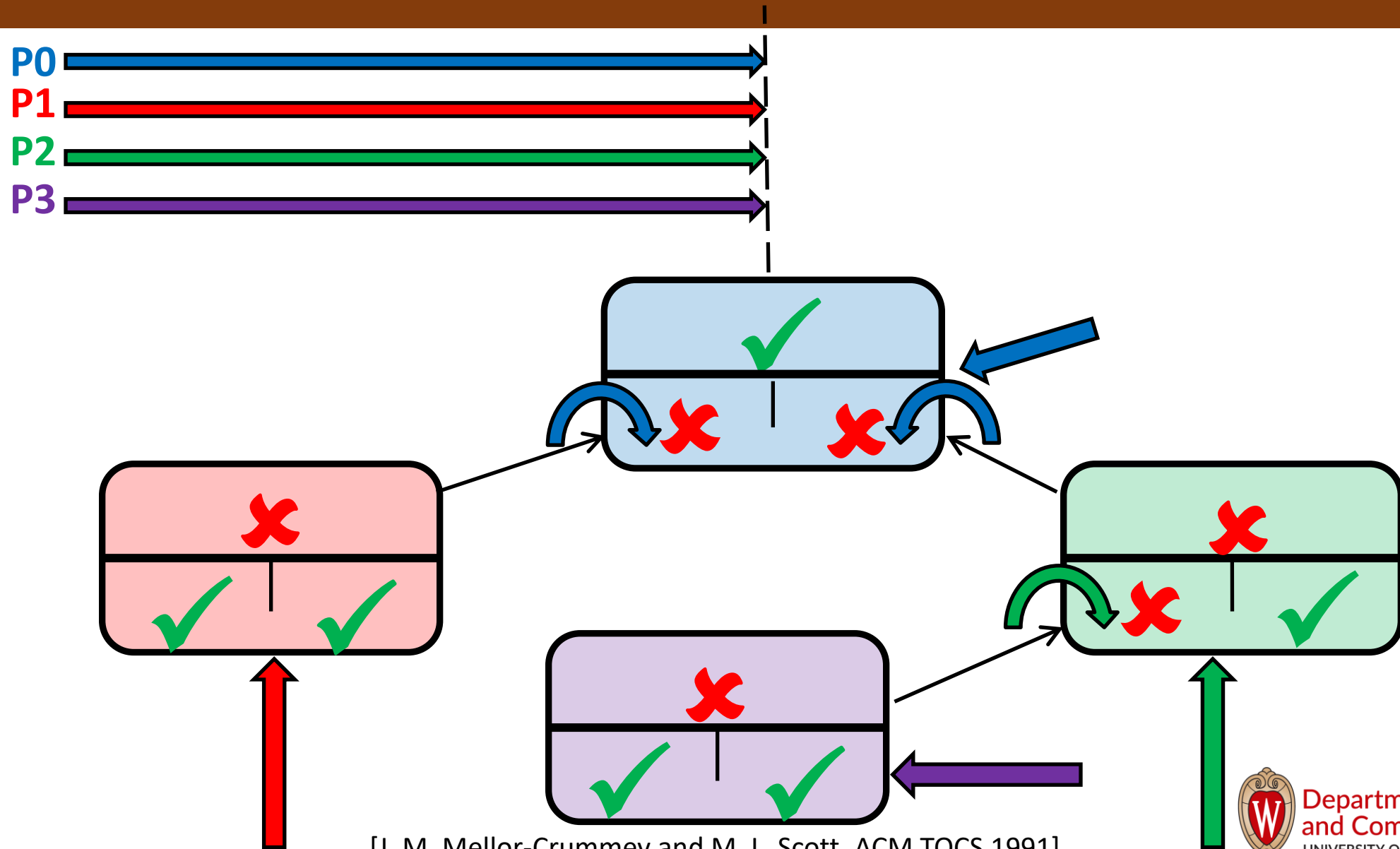
MCS Tree Barrier

P0 →
P1 →
P2 →
P3 →



[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier

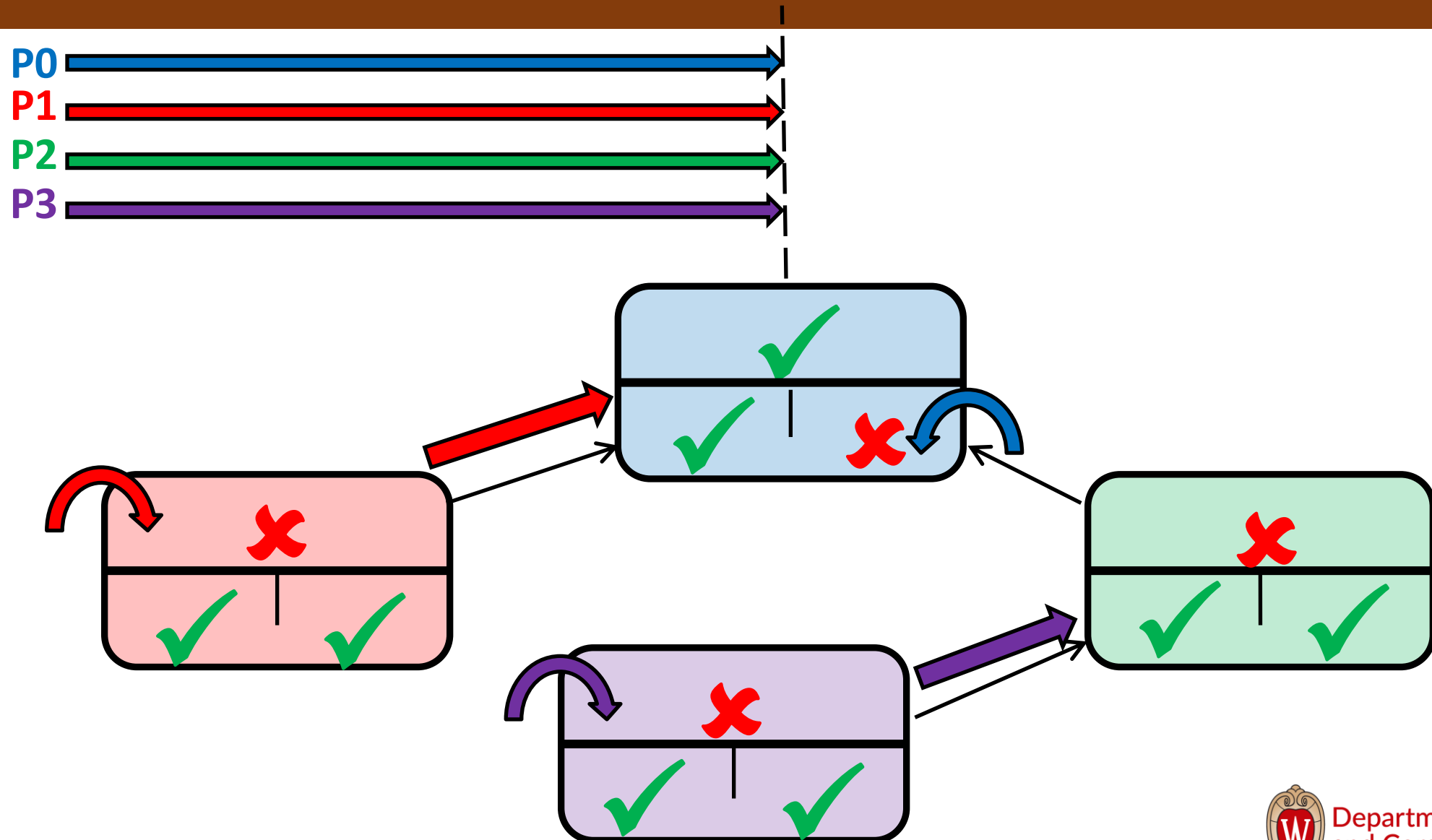


[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]



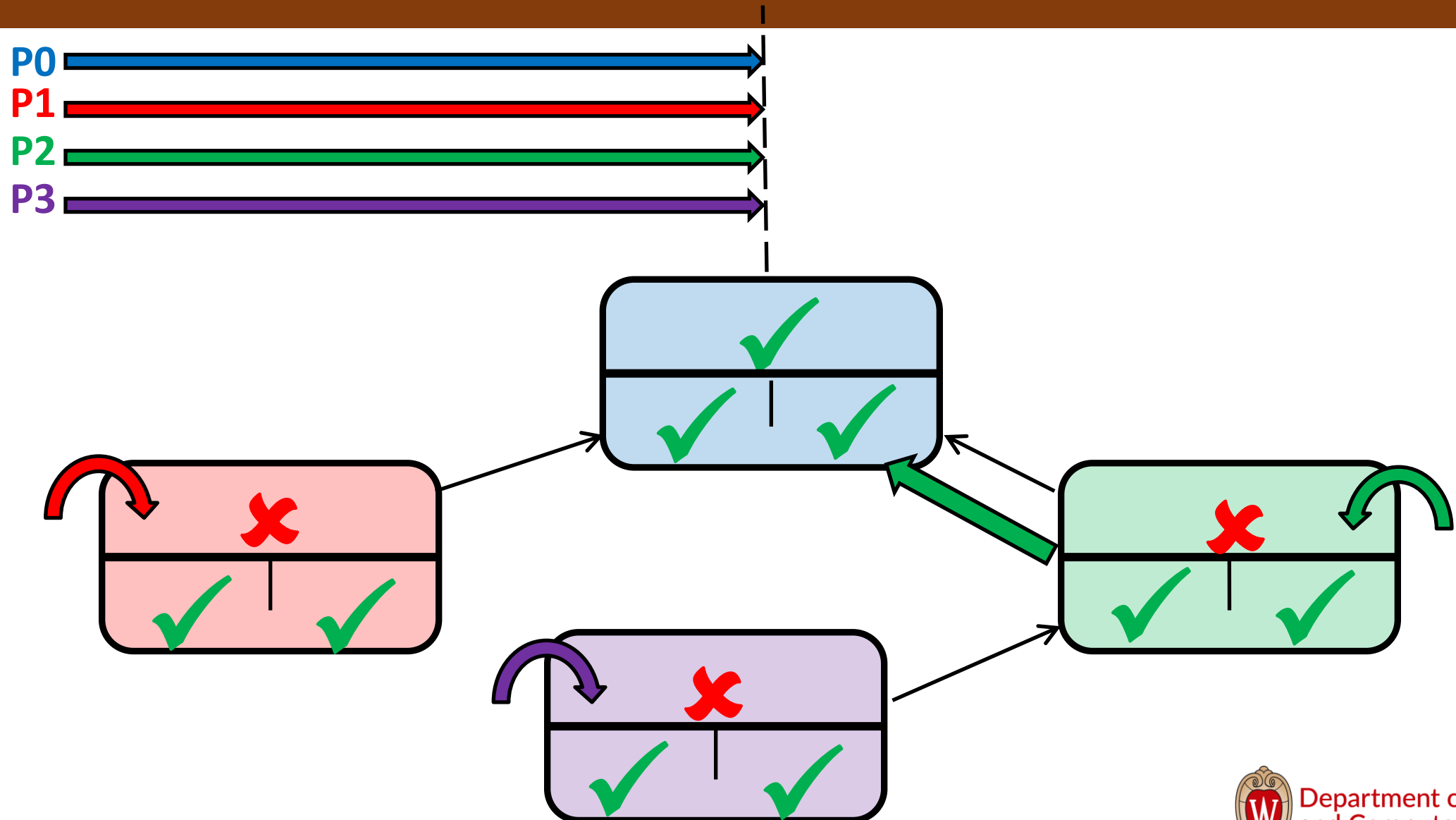
Department of Electrical
and Computer Engineering
UNIVERSITY OF WISCONSIN-MADISON

MCS Tree Barrier



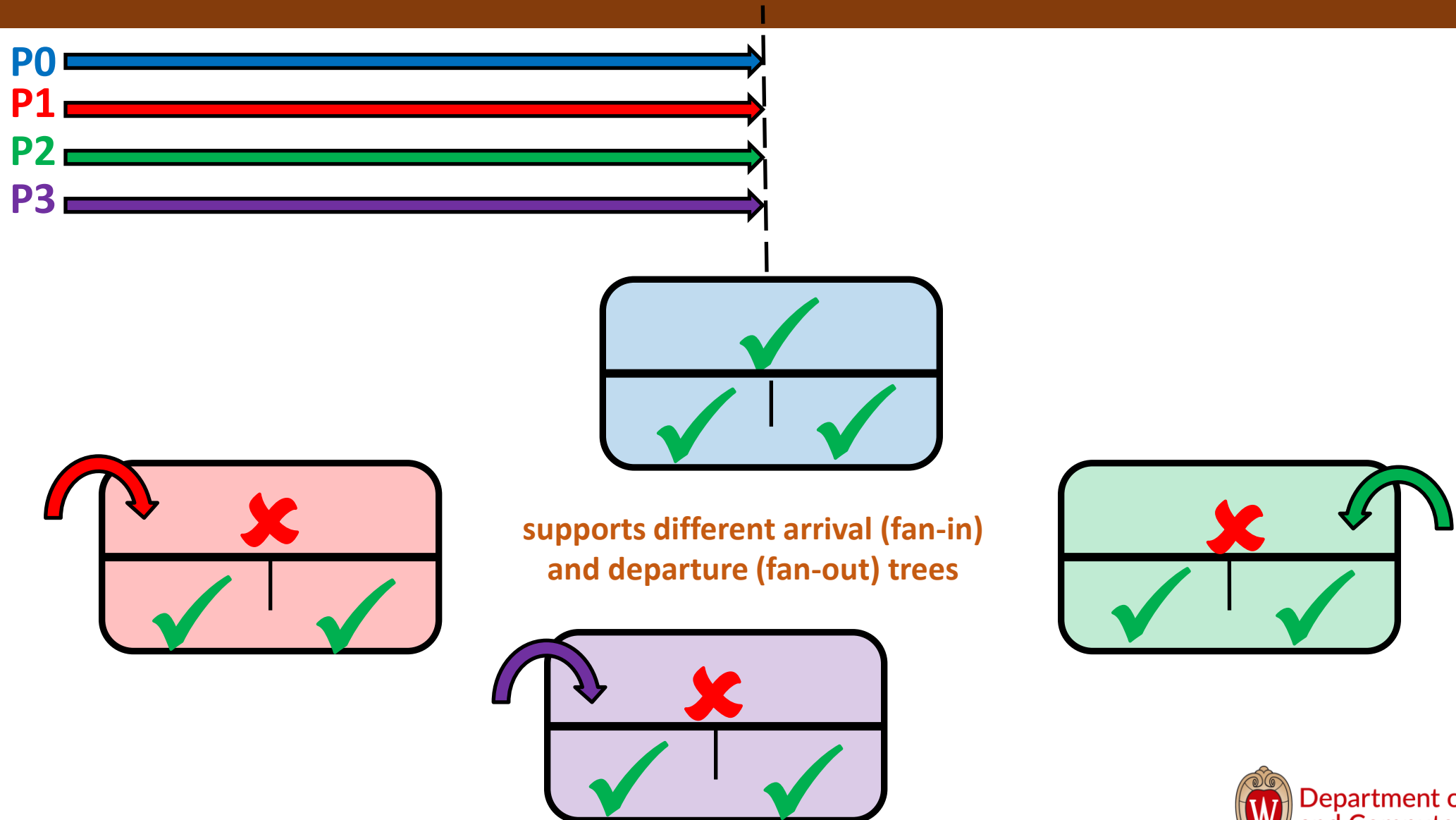
[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier



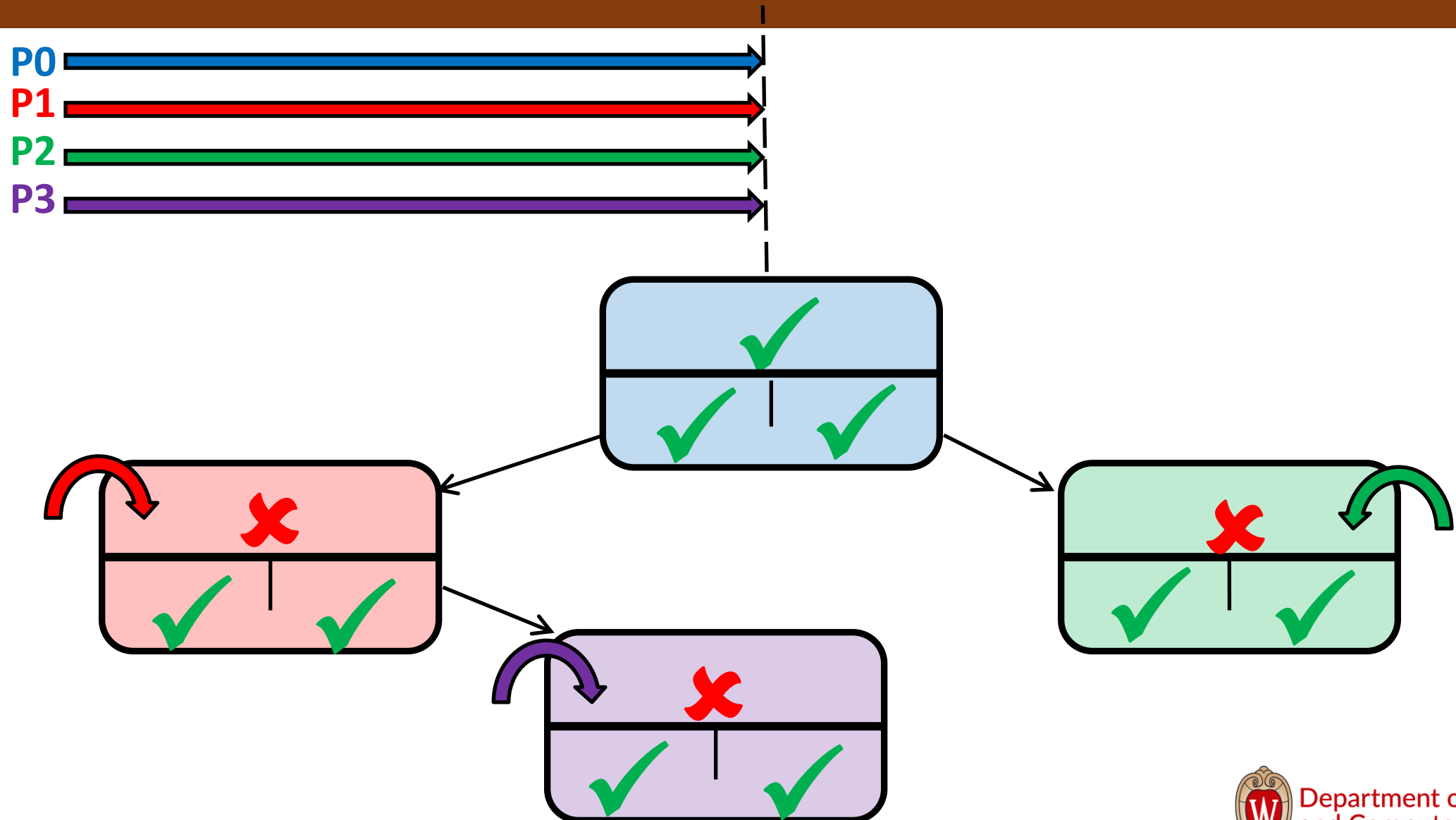
[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier



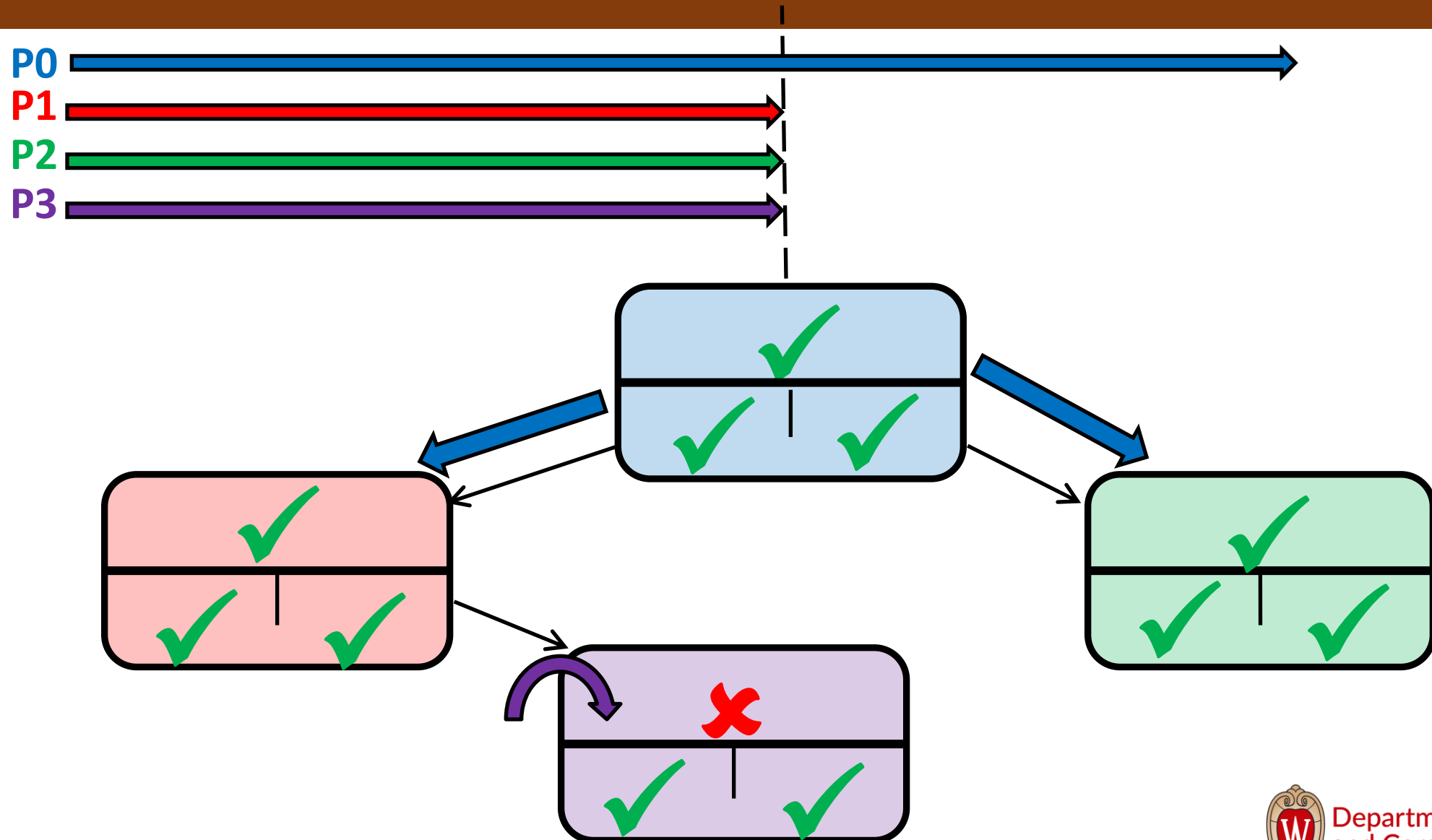
[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier



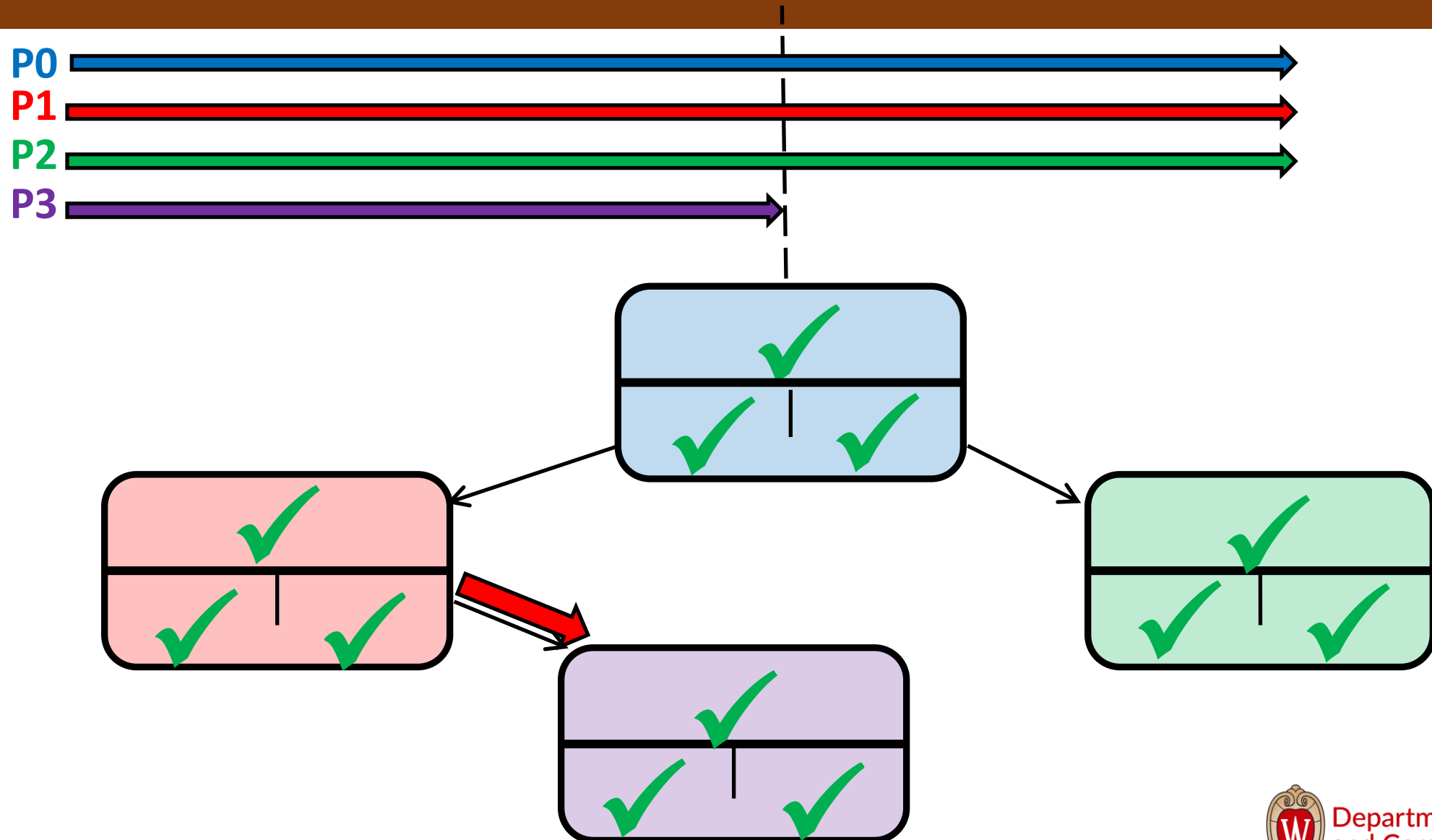
[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier



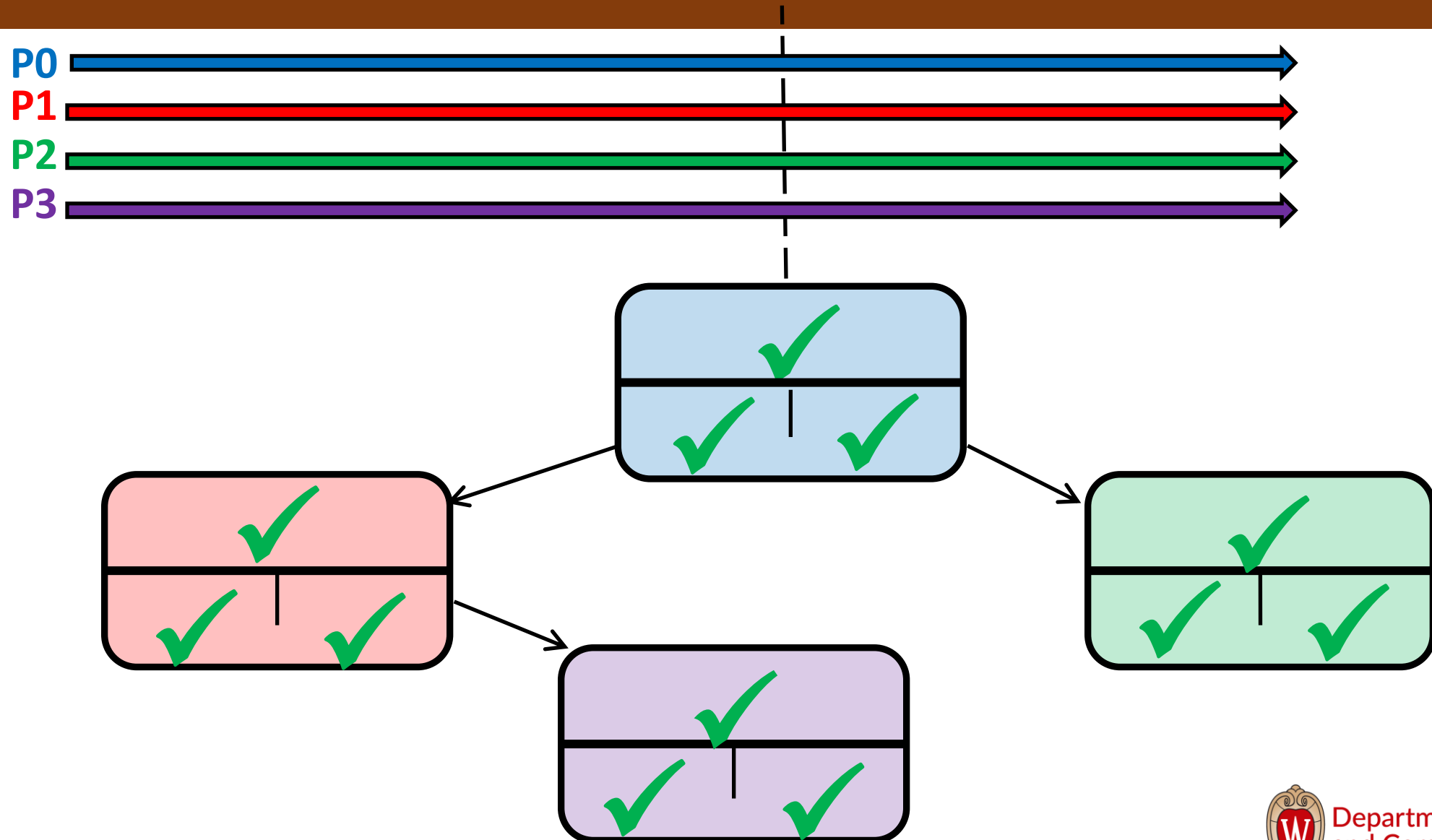
[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier



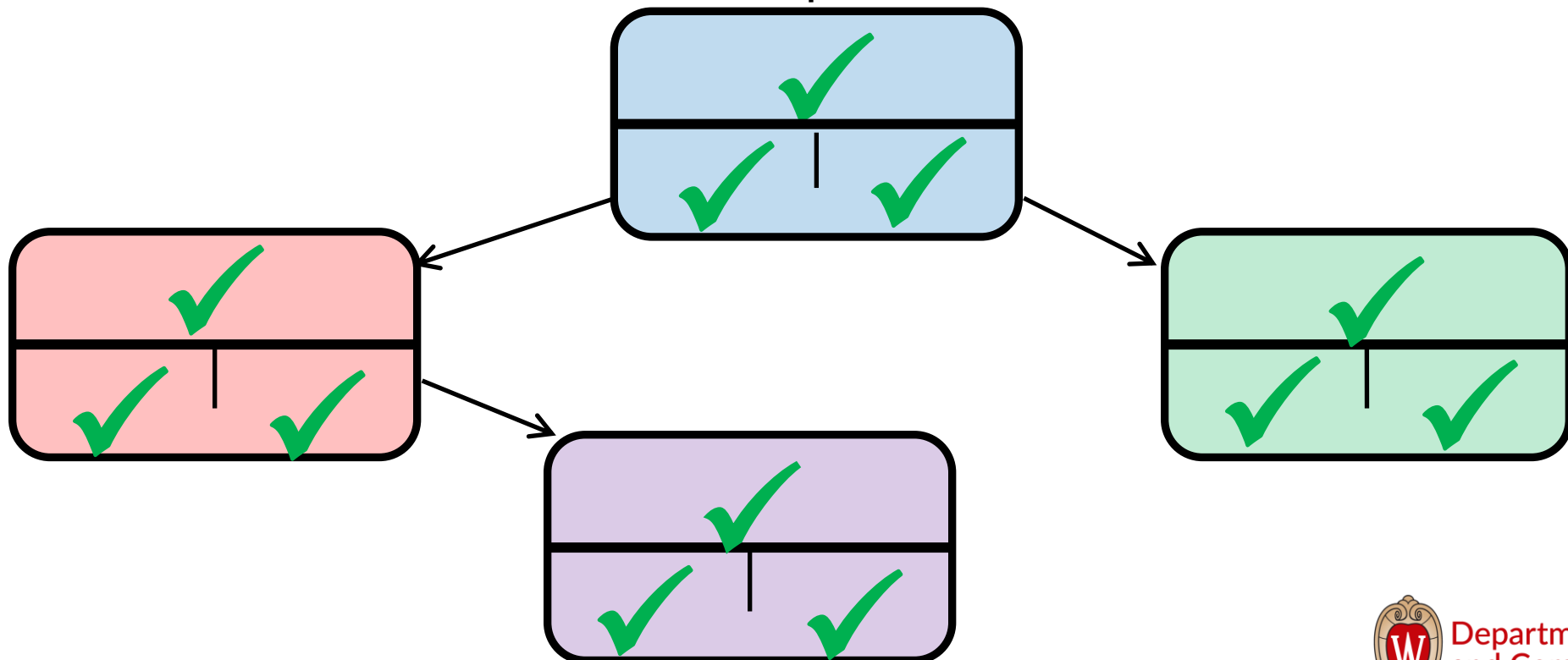
[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier



[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

MCS Tree Barrier



[J. M. Mellor-Crummey and M. L. Scott, ACM TOCS 1991]

Barriers – Summary

	centralized	SW combining	tournament	dissemination	MCS tree
private spinning	no	no	yes	yes	yes
arrival delay	$O(n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$
departure delay	$O(1)$	$O(\log n)$	$O(\log n)$	$O(1)$	$O(\log n)$
network traffic	$O(n)$	$O(n)$	$O(n)$	$O(n \log n)$	$O(n)$
storage cost	$O(1)$	$O(n)$	$O(n)$	$O(n \log n)$	$O(n)$